

ARDMS Vascular Mock Registry Exam E24

Questions first, answer key at the end. Feedback removed.

1. What is the name of the vascular device that is placed into the IVC using a catheter?

- A. Wall filter
- B. Greenfield filter
- C. Stent
- D. Pacemaker

2. A thrombus is identified in the soleal veins of the calf. The thrombus will propagate into what calf veins first?

- A. posterior tibial and peroneal
- B. peroneal, posterior tibial and anterior tibial
- C. anterior tibial
- D. great saphenous

3. Which of the following scenarios demonstrates the highest risk for embolization?

- A. 40% ICA stenosis with mild, irregular soft atheroma
- B. 60% ICA stenosis with moderate, smooth calcific atheroma
- C. Buerger disease
- D. chronic ICA occlusion

4. An arterial duplex exam demonstrates monophasic flow in the popliteal and calf arteries. The bilateral ABI calculations are over 1.0. What type of testing can be used to better evaluate the pressures in the lower extremities in this patient?

- A. Repeat the ABI using larger width of cuff to reduce the peak pressure obtained
- B. Digit plethysmography
- C. Repeat the ABI using smaller width of cuff to reduce the peak pressure obtained
- D. Impedance plethysmography

5. Which of the following signs/symptoms is an expected finding with a carotid sinus massage?

- A. hypertension
- B. dizziness
- C. tachycardia
- D. urticaria

6. Which of the following is a cause for deep venous HTN?

- A. Congestive heart failure
- B. Aortic coarctation
- C. Klippel Trenaunay Webber Syndrome
- D. Diffuse atherosclerotic disease

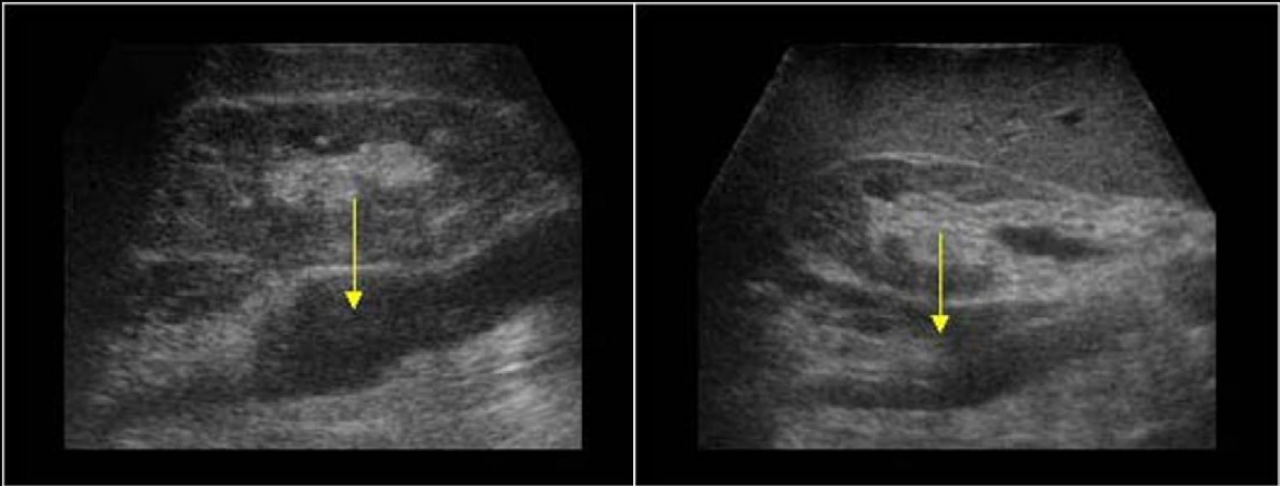
7. A pulsatility index of > 5.0 in the extremities indicates:

- A. normal resistance
- B. mildly decreased resistance
- C. moderately decreased resistance

D. severely decreased resistance

8. The 2D images displayed are from a normal renal Doppler evaluation in a patient with a history of 4.5cm distal AAA and acute onset of moderate HTN. What should you do to further evaluate or communicate the findings?

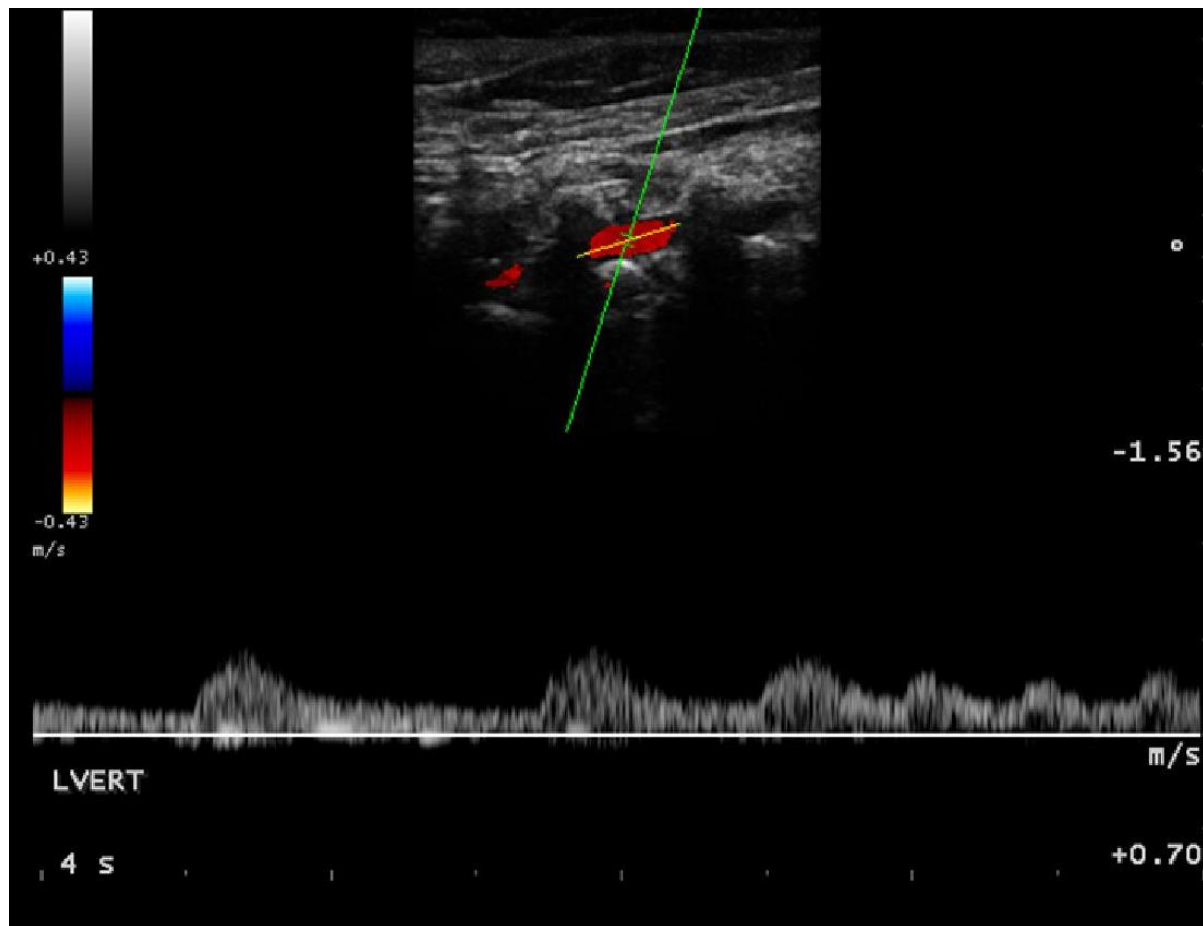
- A. Evaluate the AAA for change in size or wall continuity.
- B. Include notes regarding the presence of ascites in the peritoneal cavity in the renal artery Doppler report for the reading physician.
- C. Include notes in the renal artery Doppler report for the reading physician regarding the presence of free fluid around the kidney that is most likely due to chronic renal failure.
- D. Review the chart to confirm the suspected abnormal renal function testing results.



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

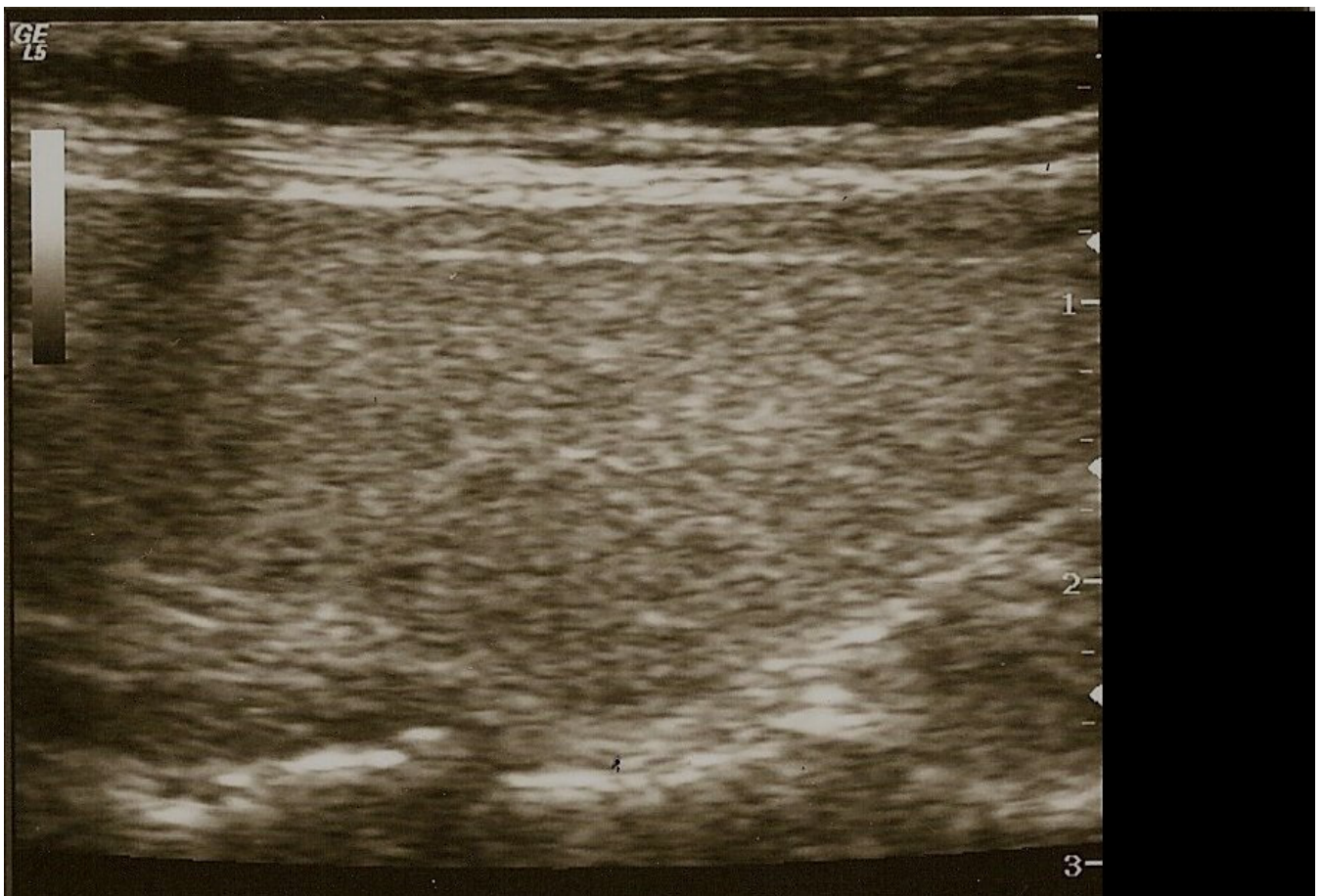
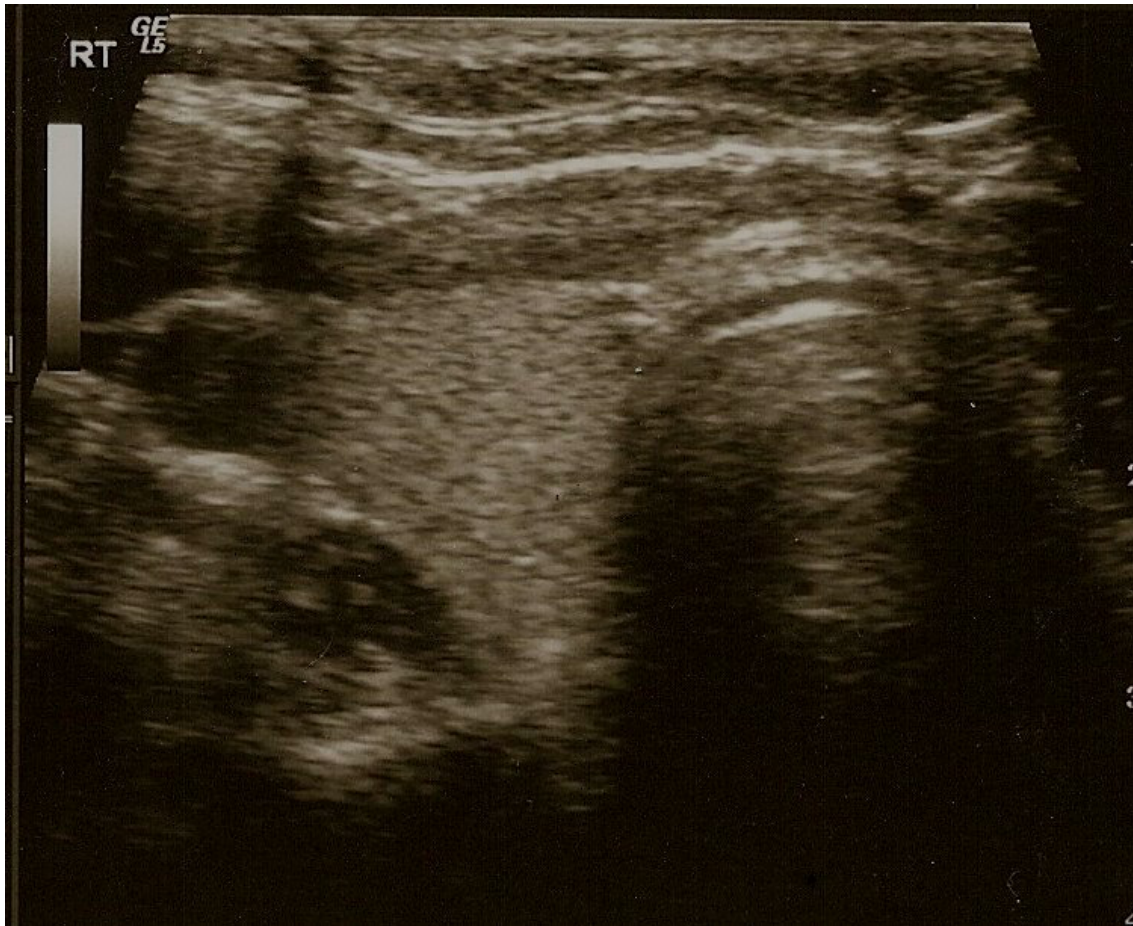
9. What is causing the waveform displayed?

- A. arrhythmia
- B. subclavian steal
- C. proximal stenosis
- D. distal stenosis



10. What is the cause of the vertical acoustic shadow seen on the images?

- A. patient motion
- B. electrical interference
- C. element damage
- D. air in the trachea



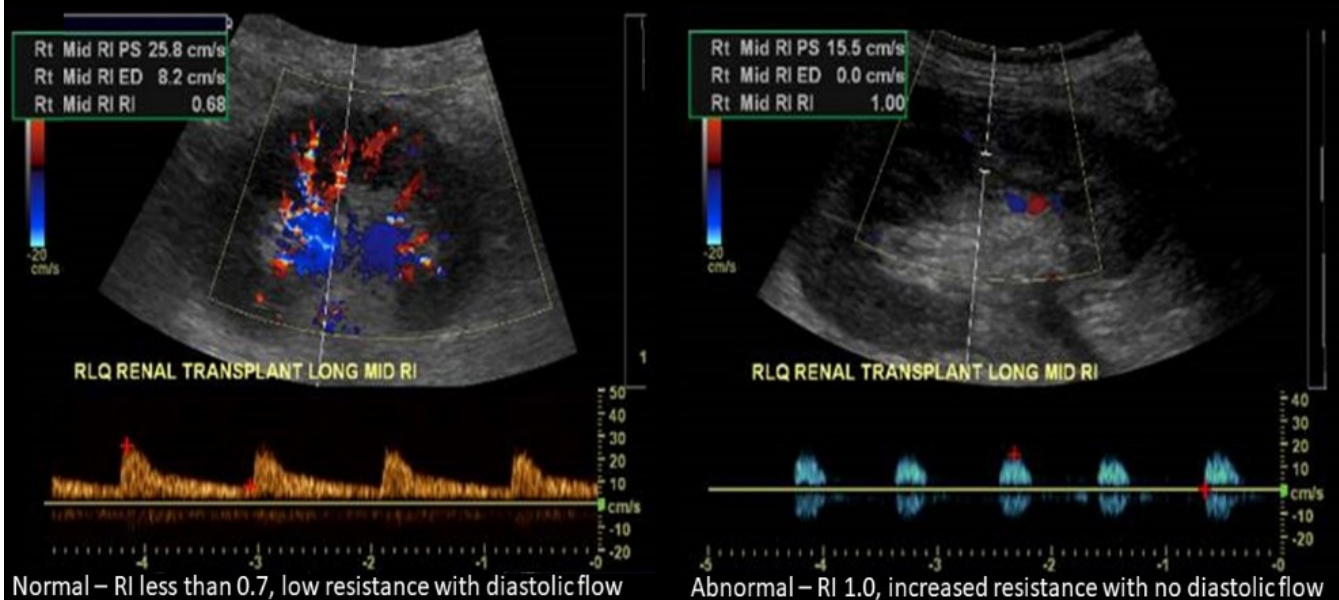
11. A patient presents for a TCD exam with a recent diagnosis of subdural hematoma. What is the physician looking for on the TCD exam?

- A. vasospasm
- B. associated intracranial stenosis
- C. associated distal carotid stenosis
- D. hematoma formation within the cranium

12. Which of the following will decrease with acute renal allograft rejection?

- A. echogenicity of the renal cortex
- B. resistive index from the arcuate arteries
- C. diastolic flow in the segmental arteries
- D. kidney size

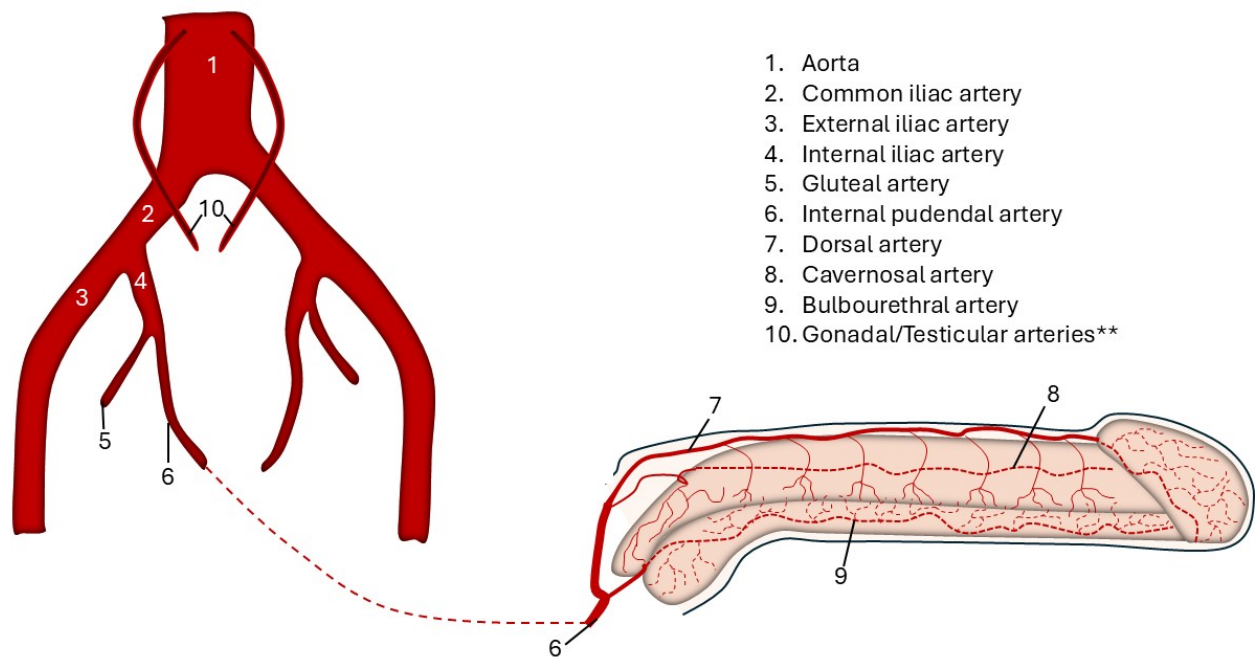
Renal Transplant Rejection



Adapted from P. Karmin, MD. <https://www.imagingrealm.com/vascular.html>

13. An abnormality in which of the following vessels could be the cause of vasculogenic impotence?

- A. external iliac and femoral arteries
- B. internal and external iliac arteries
- C. aorta and gonadal arteries
- D. common and internal iliac arteries



14. What information from an angiography report can be correlated with the ultrasound results?

- A. area stenosis
- B. flow velocity
- C. diameter stenosis
- D. areas of turbulent flow

15. A patient is referred for a Doppler ultrasound of the IVC and hepatic veins due to an abnormal echocardiogram. The report states that the hepatic veins appear to empty directly into the right atrium with limited visualization of the upper IVC. What are you looking for in the abdomen?

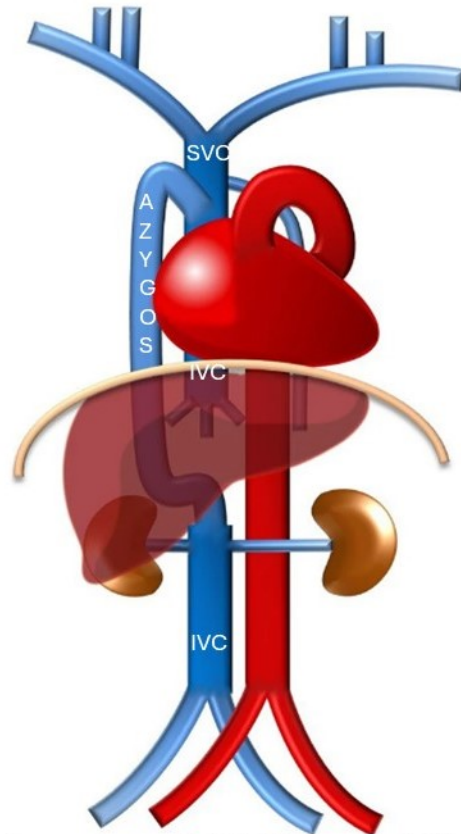
- A. azygous or hemiazygos IVC
- B. situs inversus
- C. portal HTN
- D. duplicated IVC

Note the hepatic segment of the IVC is missing.

The azygous vein offers a venous collateral pathway.

Blood moves from the distal IVC to the azygos vein and then into the SVC.

Venous blood is still returned to the right atrium.



Oliveira, J.D., Martins, I. Congenital systemic venous return anomalies to the right atrium review. Insights Imaging 10, 115 (2019). <https://doi.org/10.1186/s13244-019-0802-y>

16. What is the best Sonographic technique to use to evaluate the status of a pseudoaneurysm after 10 minutes of compression therapy is applied?

- A. CW Doppler
- B. Color Doppler
- C. PW Doppler
- D. Tissue Doppler

17. Which of the following transducers would be preferred for a lower extremity DVT evaluation on an average sized patient?

- A. 5-7MHz curvilinear
- B. 10-12MHz linear
- C. 5-7MHz linear
- D. 7-10MHz sector

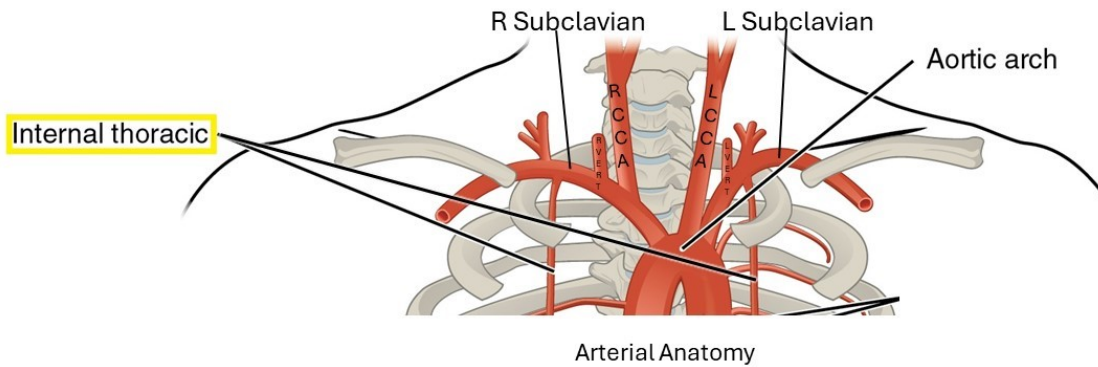
18. When evaluating a patient for placement of an AV fistula for dialysis, what is the minimum acceptable diameter for the native vein where the graft will be connected?

- A. 3.5mm
- B. 2.5mm
- C. 4.5mm
- D. 5mm

19. Which statement is true regarding the internal thoracic artery?

- A. It is a branch of the thoracic aorta.
- B. The Allen test must be performed to assess breast perfusion prior to harvesting the internal mammary artery.

- C. It can be used as a coronary bypass graft.
- D. It can be a source of collateral flow for the vertebrobasilar system.



Adapted From: Anatomy & Physiology, Connexions Web site. <http://cnx.org/content/col11496/1.6/>

20. During a color evaluation of the common femoral vein, the patient is asked to perform the Valsalva maneuver. The cephalic flow indicated by the blue color in the CFV immediately disappears until the patient releases their muscles. The blue color returns and completely fills the vessel. What do these color Doppler findings indicate?

- A. The response described is a normal result for the CFV.
- B. Superficial venous insufficiency is present.
- C. There is most likely an occlusive DVT in the superficial femoral vein causing the venous turbulence proximally.
- D. A significant level of insufficiency is present and should be further evaluated with PW or CW Doppler techniques.

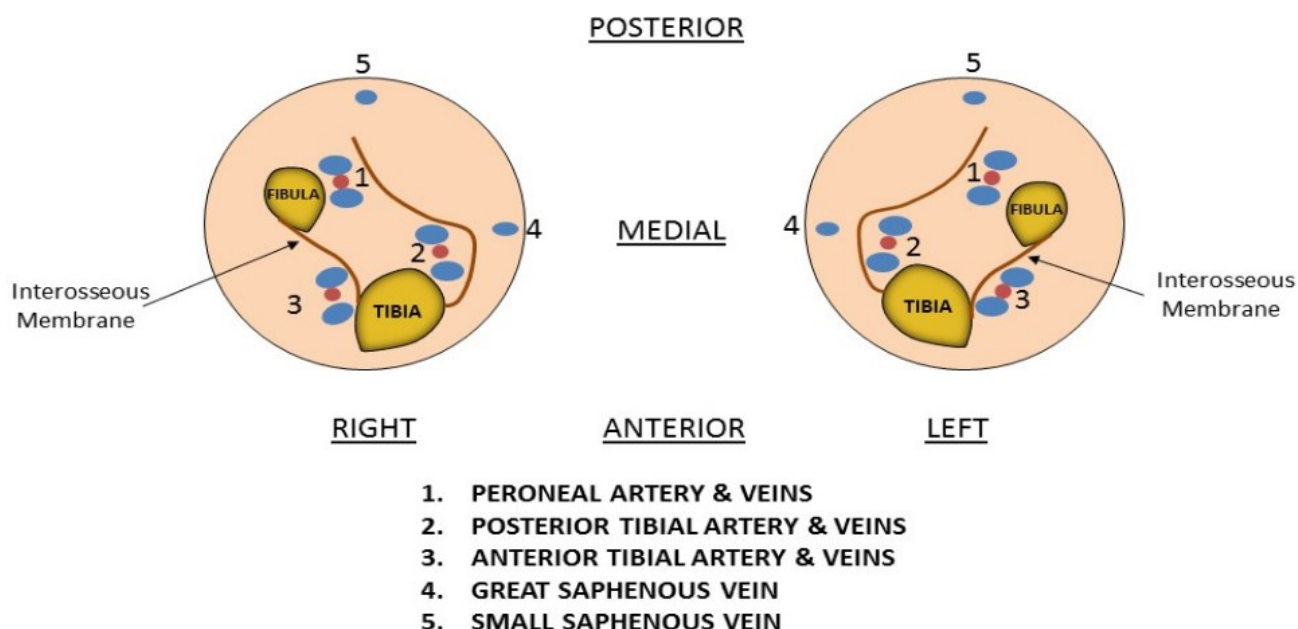
21. How does the imaging protocol differ when evaluating a hemodialysis graft versus a hemodialysis fistula?

- A. The outflow vein must be evaluated with a fistula but is not required for graft evaluation
- B. A graft has one anastomotic site and the fistula has two
- C. A graft has two anastomotic sites and the fistula only has one
- D. The outflow vein must be evaluated with a graft but is not required for fistula evaluation

22. The posterior tibial artery courses along:

- A. the medial aspect of the thigh.
- B. the medial aspect of the calf.
- C. the lateral aspect of the calf.
- D. the posterior aspect of the knee

Vascular Anatomy of the Calf



23. Which of the following explains the increased incidence of DVT in the left leg when compared to the right leg?

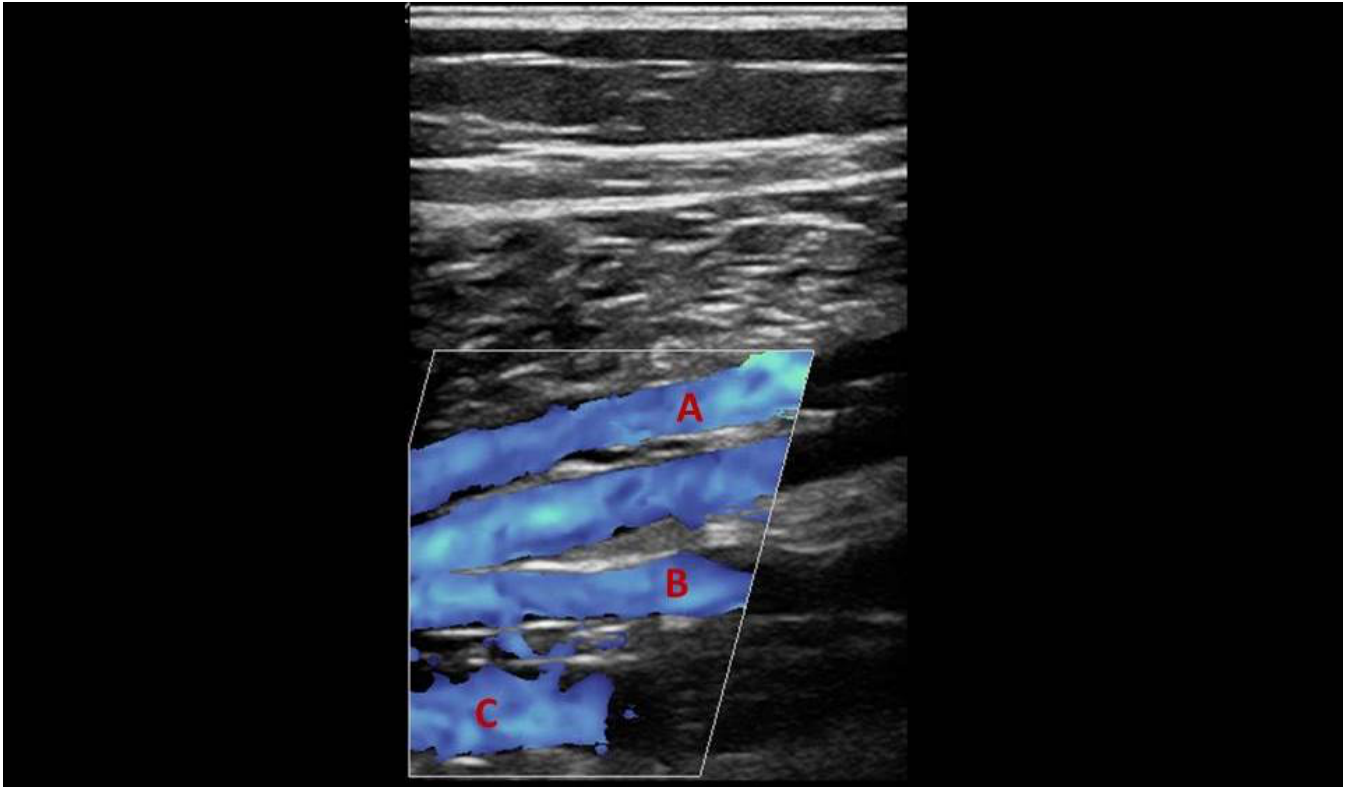
- A. Left iliac vein is a common location for DVT formation with pregnancy
- B. The left iliac vein is usually smaller than the right.
- C. There are more valves present in the left iliac vein which can lead to increased venous stasis.
- D. The anatomic course of the left iliac vein leads to increased risk of extrinsic compression.

24. A patient presents with a history of HTN, diabetes and claudication. The arterial Doppler evaluation demonstrates monophasic flow in all calf vessels, but the ABI values are LT 1.36 and RT 1.32. Which of the following correctly describes the findings?

- A. The exam is normal and additional testing should be performed to rule out extrinsic vessel compression due to popliteal entrapment syndrome.
- B. The ABI values are abnormally elevated due to a diabetic complication called medial calcification.
- C. The most likely diagnosis is thromboangiitis obliterans, which leads to medial calcification of the calf vessels.
- D. The exam is normal and additional testing should be performed to rule out extrinsic vessel compression due to median arcuate ligament syndrome.

25. The transducer is placed in the longitudinal position slightly anterior to midline of the medial upper left calf. The beam is angled posterior until the image displayed appears. What vessel is indicated by letter C?

- A. posterior tibial vein
- B. peroneal vein
- C. anterior tibial vein
- D. gastrocnemius vein



26. Upon cessation of exercise, the ankle pressures in a normal patient will:

- A. decrease to normal levels in less than 5 minutes
- B. decrease to normal levels in less than 10 minutes
- C. increase to normal levels in less than 5 minutes
- D. increase to normal levels in less than 10 minutes

27. Which of the following vascular complications is treated with a fasciotomy?

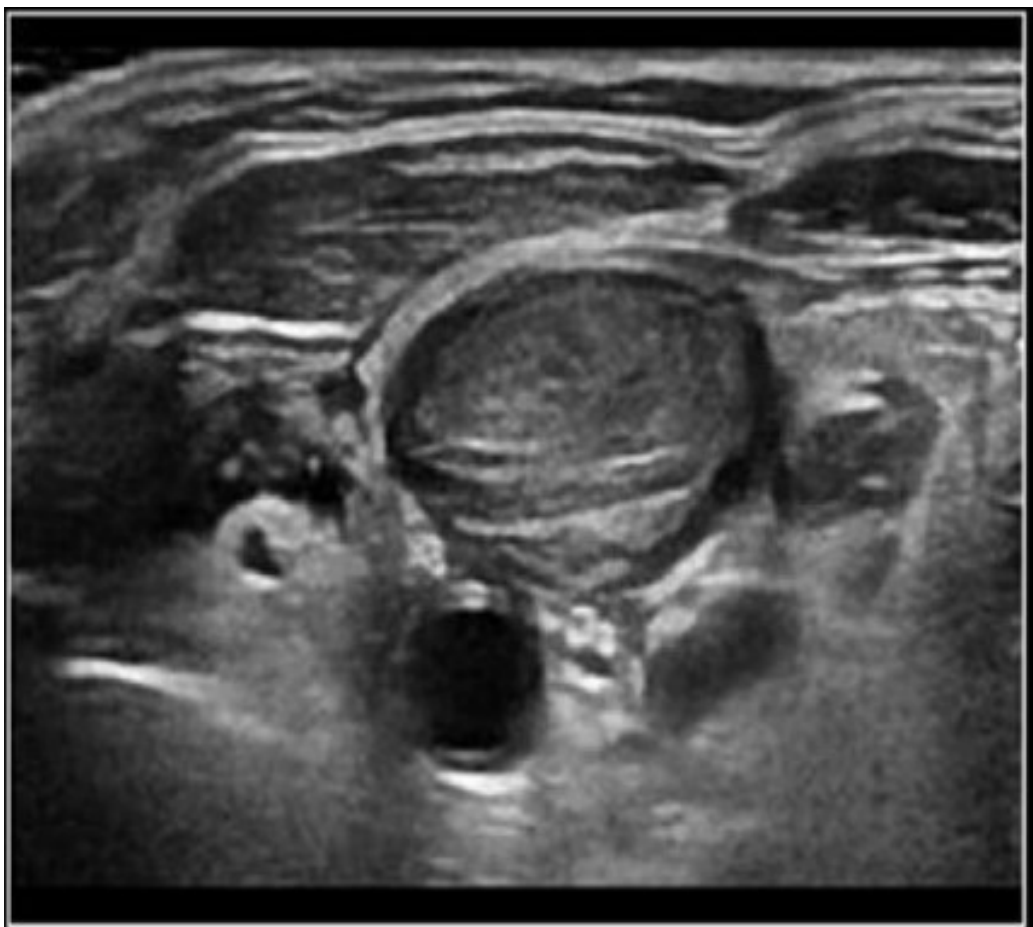
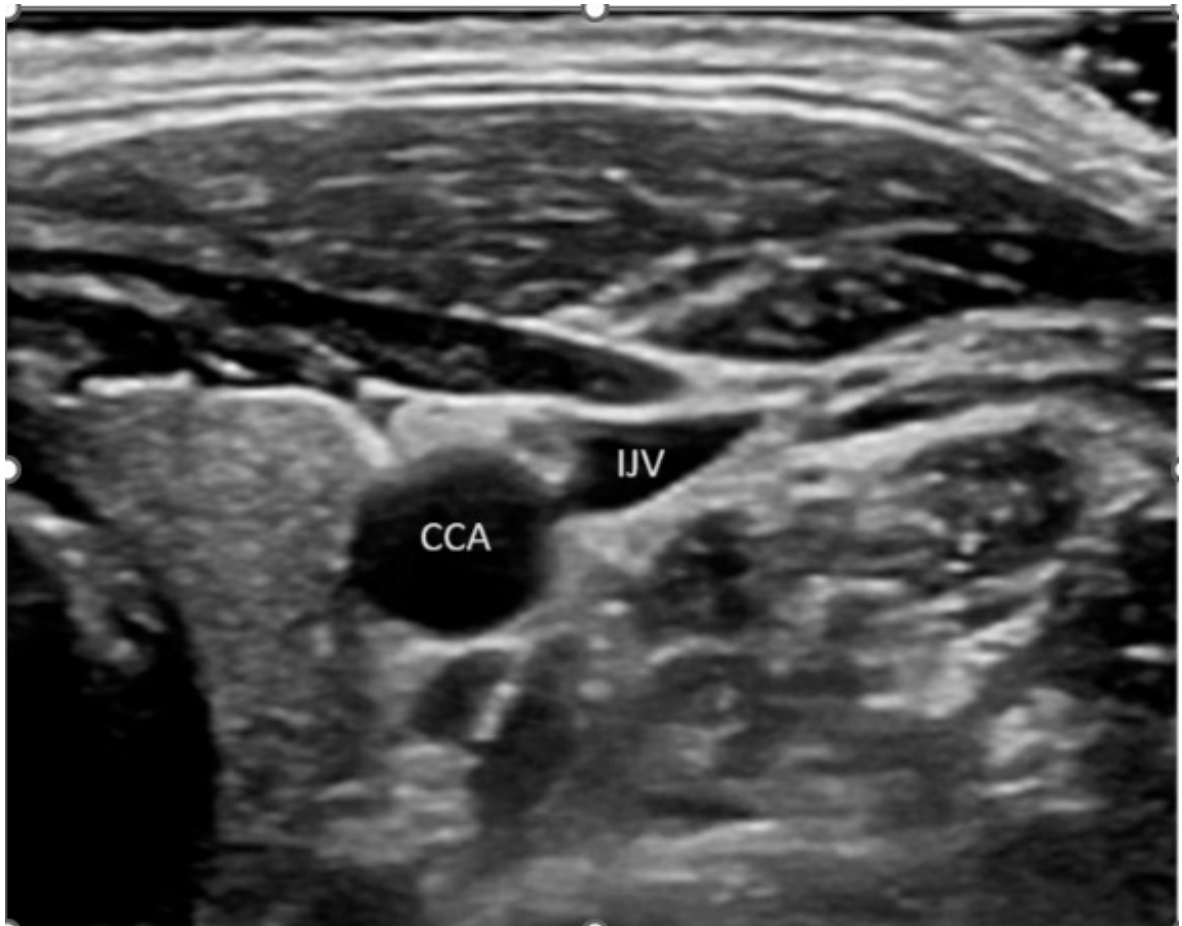
- A. anterior compartment syndrome
- B. pseudoaneurysm
- C. dissection
- D. popliteal entrapment

28. Which of the following symptoms would be associated with steal syndrome of a hemodialysis graft?

- A. hand pain that increases with exercise
- B. hand pain that is reduced or eliminated during dialysis
- C. spontaneous bruising of the neck, arm and hand
- D. shoulder swelling

29. Which of the following is a sonographic finding of acute DVT?

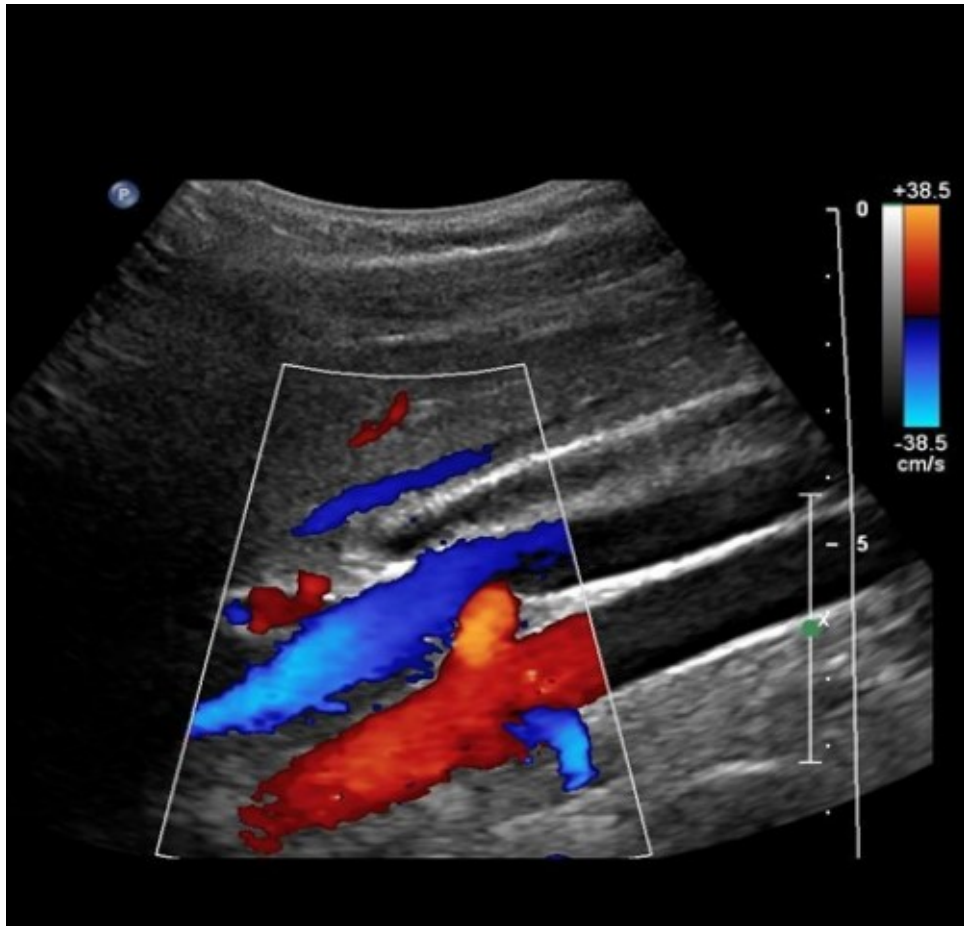
- A. Partial compressibility of the vein
- B. Flattened, collapsed vessel with increased echogenicity of the wall
- C. Valvular insufficiency
- D. Engorged vein with increased echogenicity in the lumen



30. Which imaging plane can produce a longitudinal image of the IVC and the aorta on the same image?

- A. mid sagittal
- B. coronal and transverse
- C. coronal
- D. transverse

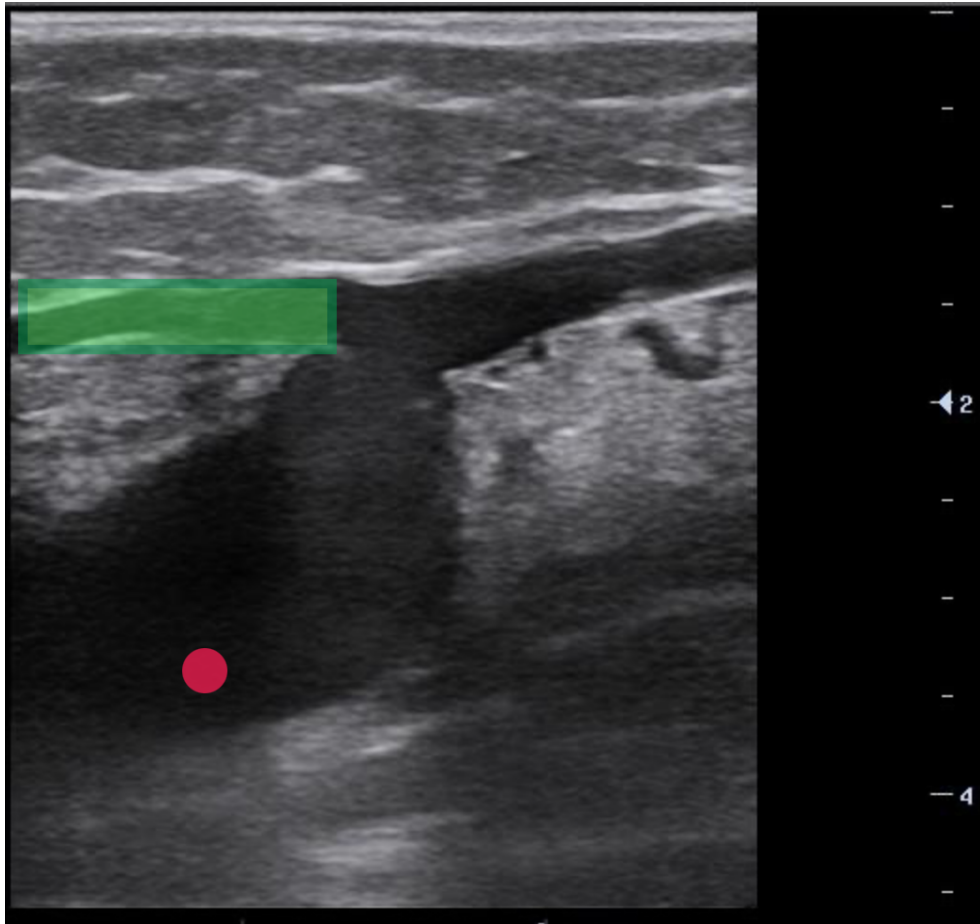




31. The Doppler tracing demonstrates electrical noise. How can you reduce or eliminate the artifact?

- A. Extend the cord to move the machine as far from the electrical outlet as possible
- B. Decrease wall filter settings
- C. Decrease the PRF
- D. Increase the wall filter settings

32. The image displayed is from the right groin in a patient with suspected venous insufficiency. Use your mouse to position the cursor over the superficial epigastric vein and click to select it.



33. The supraorbital artery is a branch of the _____. Branches of the supraorbital artery connect with branches of the _____, which is a branch of the ECA. This is a common collateral pathway from the ECA to the ICA.

- A. ophthalmic artery, superficial temporal artery
- B. ophthalmic artery, maxillary artery
- C. ICA, facial artery
- D. ICA, ophthalmic artery

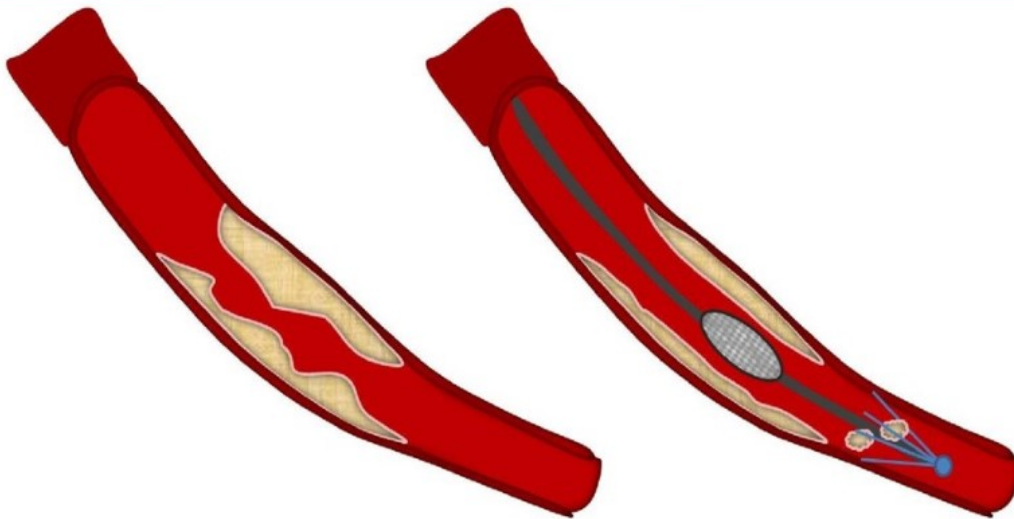
34. Why is the carotid compression technique performed?

- A. to evaluate the presence of intracranial collateral formation
- B. to evaluate the carotid as a potential arterial graft
- C. to evaluate the effects of vasospasm
- D. to evaluate the venous outflow of the intracranial vessels

35. _____ uses a catheter with a rotating head to break up the plaque and suction capabilities to remove it from the artery.

- A. An angioplasty procedure
- B. An endarterectomy procedure
- C. An atherectomy procedure
- D. A valvulotome procedure

Atherectomy



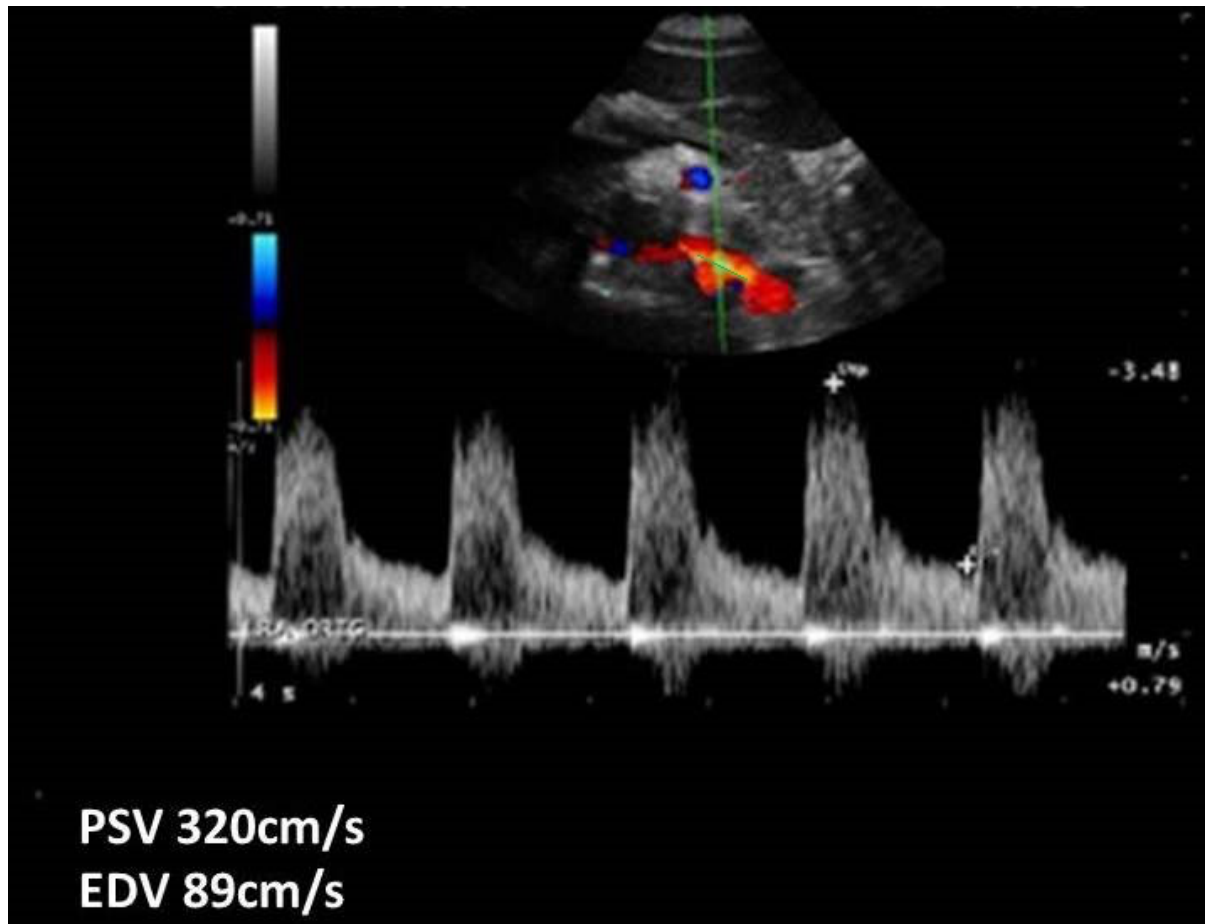
A catheter with a swivel head scraping attachment is advanced into the area of stenosis within the artery. The catheter head is rotated to break up the plaque and an umbrella-like tip catches the atheroma pieces for retrieval from the artery.

36. Which of the following correctly describes the sensitivity of a testing technique?

- A. The ability of a test to detect disease when it IS present.
- B. If the test result is negative sensitivity is the probability that the patient does not have disease
- C. If the test result is positive, sensitivity is the probability that the patient actually has the disease
- D. The ability of a test to rule out disease when it IS NOT present.

37. Which of the following statements is true regarding the Doppler tracing from the renal artery?

- A. Normal flow pattern and velocity are demonstrated
- B. According to RAS diagnostic criteria, Stenosis > 60% is demonstrated
- C. The renal to aortic ratio is most likely below 3.5
- D. Normal flow pattern and velocity are demonstrated but an incorrect Doppler angle is used to assess the flow causing aliasing to occur



38. The renal-aortic ratio is invalid in patients with:

- A. diabetes
- B. abdominal aortic aneurysm
- C. medullary sponge kidney
- D. renal vein thrombosis

39. Which of the following requires a 2D and Doppler evaluation for a complete exam?

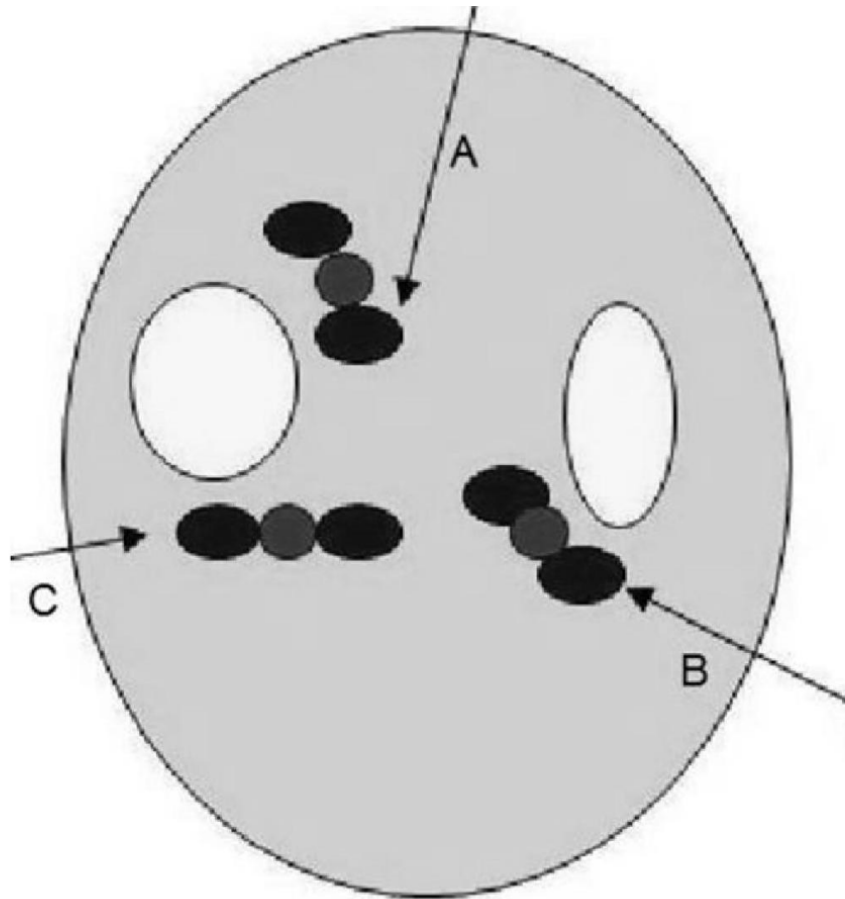
- A. Raynaud syndrome
- B. Popliteal entrapment
- C. Thoracic outlet syndrome
- D. Renal or liver transplant

40. Which of the following are common risk factors for varicose vein formation?

- A. post-phlebitic syndrome and high cholesterol
- B. diabetes and smoking
- C. smoking and high cholesterol
- D. pregnancy and standing for long periods of time

41. What blood vessels are located at letter A?

- A. 1 PTA and 2 PTV
- B. 2 PTA and 1 PTV
- C. 2 ATA and 1 ATV
- D. 1 ATA and 2 ATV



42. At the start of an endovenous ablation procedure for the great saphenous vein, the catheter is inserted _____ and advanced to a position _____ before activating the device.

- A. into the distal GSV, 2cm distal to the saphenopopliteal junction
- B. into the distal GSV, 2cm distal to the saphenofemoral junction
- C. into the GSV at the groin, at the most distal point of reflux
- D. into the GSV at the groin, 2cm below the knee

43. According to the findings on this arteriogram, which vessel will demonstrate a post-stenotic flow pattern on a carotid Duplex exam?

- A. Proximal L CCA
- B. R and L vertebral arteries and R Subclavian
- C. RCCA, R Vertebral, R Subclavian
- D. L Vertebral and L Subclavian



44. A normal lower extremity arterial response to reactive hyperemia is:

- A. <50% increase in ankle pressures with return to baseline in under 5 minutes
- B. no change in ankle pressures
- C. <35% decrease in ankle pressures that return to baseline within one minute
- D. <50% decrease in ankle pressures that return to baseline within one minute

45. Where is the marginal artery of Drummond located?

- A. part of the cerebrovascular circulatory system
- B. at the splenic hilum
- C. connects the SMA and IMA through the mesentery
- D. connect the dorsalis pedis artery to the posterior tibial artery in the foot

46. What cardiac complication is related to the appearance of the carotid waveform?

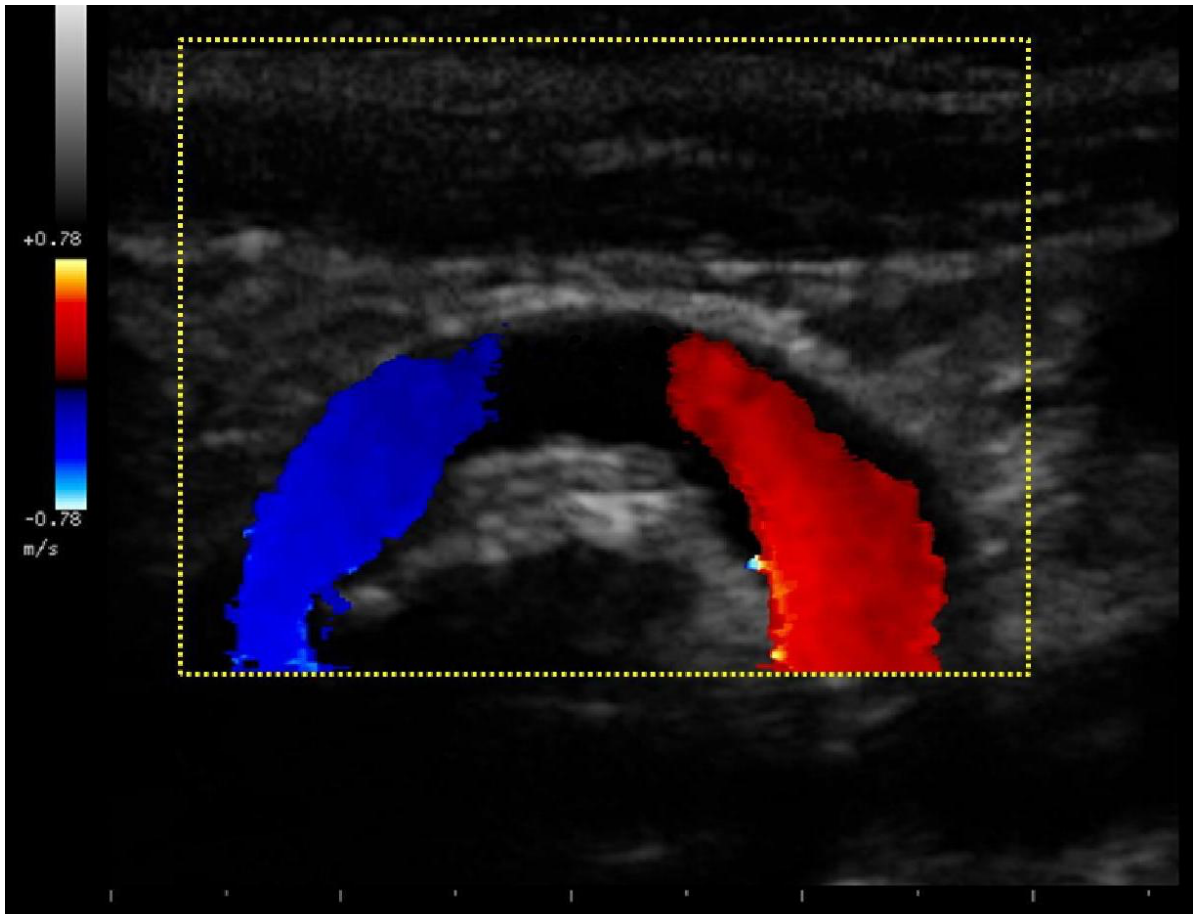
- A. aortic regurgitation
- B. mitral regurgitation
- C. pulmonary stenosis
- D. cardiac tamponade

47. What is the normal value for a transcutaneous oximetry exam?

- A. <40mmHg
- B. 2.0 (unitless)
- C. 70-80mmHg
- D. 3.5 (unitless)

48. Which of the following statements best describes the reason why there is an absence of color flow in the vessel on the image?

- A. The ultrasound beam is parallel to the motion of the blood, so no flow is detected.
- B. The ultrasound beam is perpendicular to the motion of the blood, so no flow is detected.
- C. The color Doppler gain is set too low to detect the slower flow at the curve in the vessel.
- D. There is an occlusion present. Flow is demonstrated proximal to the lesion with collateral flow filling the distal portion of the vessel.



49. Which of the following would not result in a carotid bruit?

- A. 60% ECA stenosis
- B. 75% ICA stenosis
- C. 99% ICA stenosis
- D. aortic valve stenosis

50. According to the Appropriate Use Criteria, which of the following is an appropriate indication for a renal artery duplex exam?

- A. Unilateral renal enlargement
- B. Peripheral arterial disease with well controlled hypertension
- C. Bilateral renal enlargement
- D. Hypertension in a young person (age <35 years)

Appropriate Use Criteria – Renal Artery Duplex Exam

ALWAYS APPROPRIATE

- Hypertension in young person (age <35 years)
- Resistant hypertension (>140/90 mm Hg on ≥3 meds)
- Malignant hypertension (>160/80 mm Hg)
- Blood pressure becoming less manageable over time
- Unexplained size discrepancy between kidneys
- Unknown cause of azotemia
- Elevated creatinine levels
- Epigastric bruit
- Acute renal failure with aortic dissection

Appropriate Use Criteria – Renal Artery Duplex Exam

ALWAYS APPROPRIATE

- Refractory heart failure
- (Symptoms of CHF at rest or even with optimal drug therapy)
- “Flash” pulmonary edema
- (Respiratory distress due to the rapid accumulation of fluid within the lung tissues with heart failure)
- Follow up known renal artery stenosis
- 1 month follow up baseline exam post-revascularization
- Follow up exam post-revascularization with new or worsening symptoms
- Follow up exam every 12 months in stable patient

INAPPROPRIATE

- Atherosclerotic vascular disease in other areas (e.g., PAD, CAD) and well controlled hypertension

51. According to the Appropriate Use Criteria, which of the following is an inappropriate indication for an upper extremity arterial duplex exam?

A. unilateral cold hand

- B. 1 month follow up baseline exam post-intervention
- C. unilateral arm swelling
- D. suspected positional arterial obstruction

52. _____ is a nonspecific sign of DVT. It is defined as pain experienced upon quick dorsiflexion of the foot.

- A. Henderson Sign
- B. Homan Sign
- C. Asher Sign
- D. Allen Sign

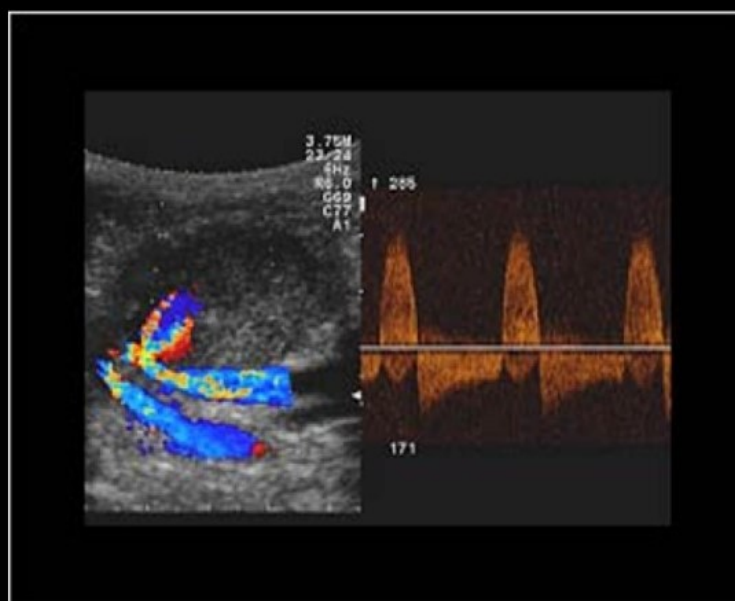
53. The resistive index of the renal parenchymal arteries in cases of main renal artery stenosis will _____ and the resistive index of the renal parenchymal arteries in cases of renal parenchymal disease will _____

- A. increase, remain normal
- B. decrease, increase
- C. remain normal, decrease
- D. increase, decrease

54. Which of the following statements best describes the Doppler tracing found in the stalk of a pseudoaneurysm?

- A. phasic forward flow
- B. low resistance to-and-fro flow
- C. continuous forward flow
- D. high resistance to-and-fro flow

Pseudoaneurysm



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

55. Saphenous vein mapping is used to assess the vein for a potential bypass conduit. Which of the following characteristics would still allow the surgeon to use the vein?

- A. lumen <2mm
- B. thrombosis
- C. valvular damage
- D. chronic obstruction of the ipsilateral deep system

56. Which of the following is an indirect sign of an unstable AAA sac with an endoleak?

- A. Echolucent areas within the sac
- B. Reduction in aneurysm shape and/or sac size
- C. Absence of pulsatility of the sac
- D. Triphasic flow in the iliac arteries

57. Warfarin is administered to a patient:

- A. to assist in atheroma lysis and recanalization
- B. to prevent thrombus propagation
- C. to assist in clot lysis and recanalization
- D. to increase platelet aggregation

58. Which of the following could cause a false reduction in ankle pressure on an average sized patient?

- A. using the 12cm cuff for the ankle
- B. medial calcification of the calf vessels
- C. cuff is loosely wrapped around the arm
- D. evaluating the pressure with the patient seated in a chair

59. Which of the following are common symptoms of chronic venous disease?

- A. brawny discoloration and pitting edema
- B. pallor and paresthesia
- C. pitting edema and ulceration on great toe
- D. varicosities and pallor

60. Which of the following measurements is used to differentiate inflow from outflow disease in the lower extremity arteries?

- A. Resistive index
- B. S/D ratio
- C. Pulsatility index
- D. Acceleration time

61. The internal iliac artery:

- A. courses lateral to the external iliac artery to reach the pelvis.
- B. branches into the superficial and deep femoral arteries
- C. demonstrates lower resistance flow than the external iliac artery.
- D. supplies the bladder, prostate and testicles in a male patient.

62. A diabetic patient presents for a segmental pressure exam. The patient complains of an injury to his big toe that won't seem to heal and has been open for over 3 weeks. The digital pressure in the affected toe will most likely be _____.

- A. >200mmHg

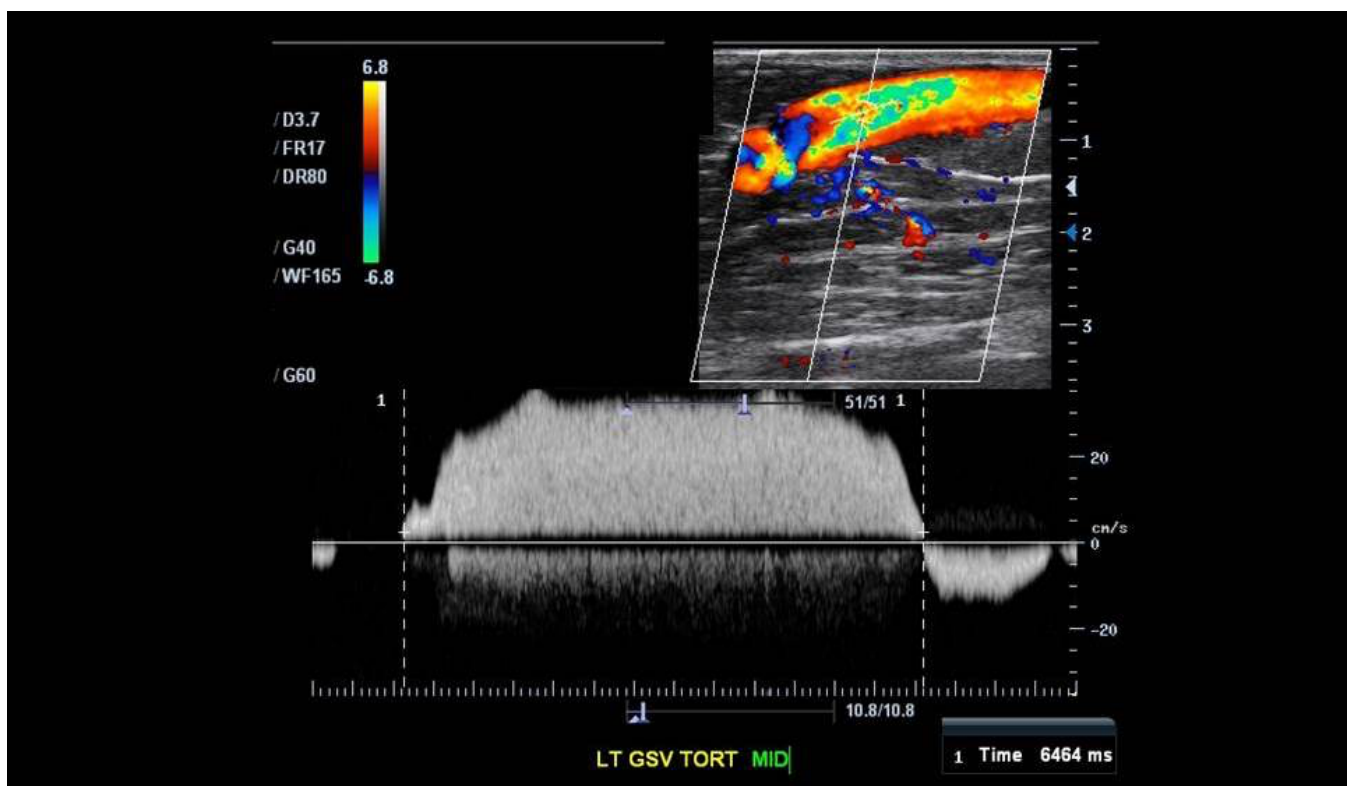
- B. 30-60mmHg
- C. 60-90mmHg
- D. <30mmHg

63. Why is an acute arterial occlusion in the lower extremity considered a critical finding?

- A. collaterals have not had time to form to redistribute the blood
- B. the risk of arterial rupture is significantly increased
- C. because it is not treatable
- D. the risk of stroke is significantly increased

64. Which of the following would be contraindicated in this patient?

- A. compression stockings
- B. daily exercise
- C. thermal ablation
- D. Coumadin therapy



65. Adventitial cysts:

- A. are usually treated with bypass graft placement
- B. most commonly affect the popliteal artery
- C. are a normal variant that can be seen in up to 10% of patients
- D. are more common in women

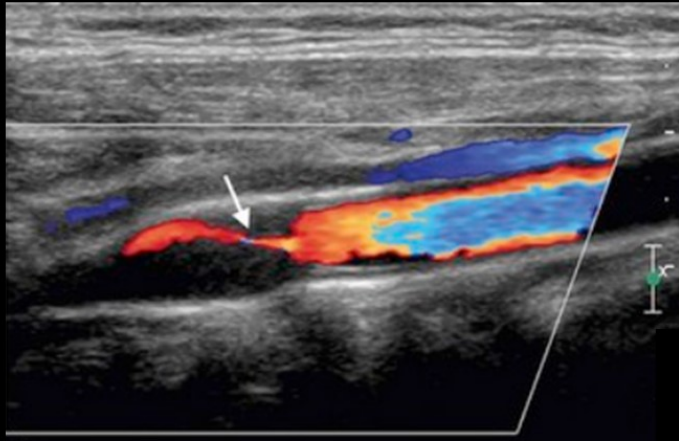
66. What is the normal change in flow velocity in response to releasing the cuff during reactive hyperemia?

- A. decrease by more than 35% below the resting flow velocity
- B. increase by more than 35% above the resting flow velocity
- C. increase to more than double the resting flow velocity
- D. decrease to less than half of the resting flow velocity

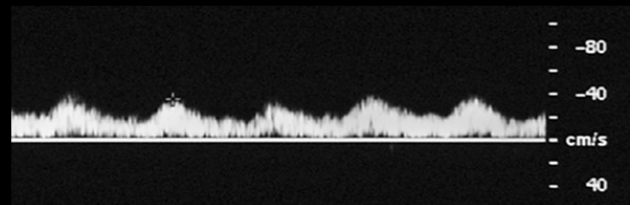
67. The string flow sign is associated with which of the following diagnoses?

- A. superficial phlebitis
- B. fibromuscular dysplasia
- C. AV fistula
- D. critical ICA stenosis

String Flow/Trickle Flow



String or trickle flow is the minimal amount of flow in a nearly occluded vessel. It will demonstrate a blunted, low velocity waveform.



68. Which of the following describes the 2D appearance of acute renal allograft rejection on ultrasound?

- A. Increased echogenicity and size
- B. Decreased echogenicity and size
- C. Resistive index over 0.8
- D. Increased echogenicity and decreased size

Renal Transplant

Normal Transplant Kidney



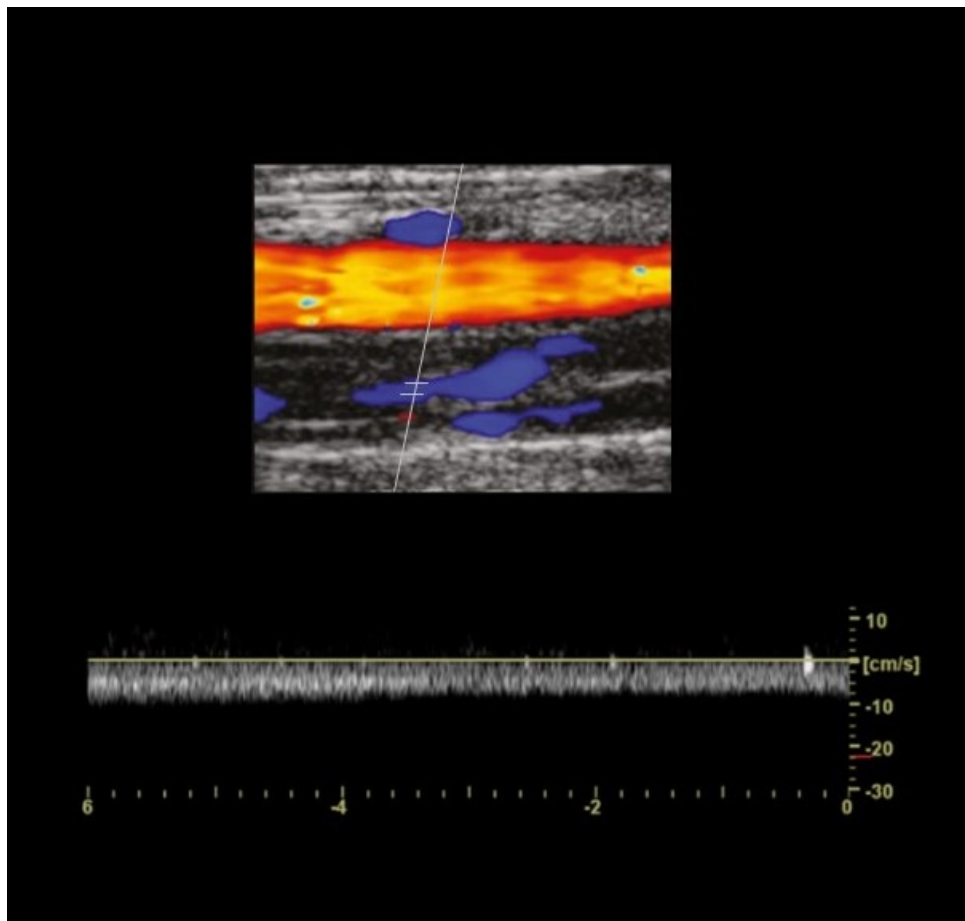
Acute Rejection – Enlarged kidney with increased echogenicity and loss of corticomedullary definition



Kurkowski S C, Thimmesch M J, Abdelghani A, et al. (June 23, 2022) A Case of De Novomembranous Nephropathy Causing Renal Transplant Rejection. Cureus 14(6): e26246. doi10.7759/cureus.26246

69. Which of the following statements is true regarding a partial venous obstruction?

- A. May cause continuous flow
- B. May cause phasic flow
- C. Usually results in significant flow reversal during Valsalva maneuver
- D. May cause increased volume of flow seen with distal augmentation



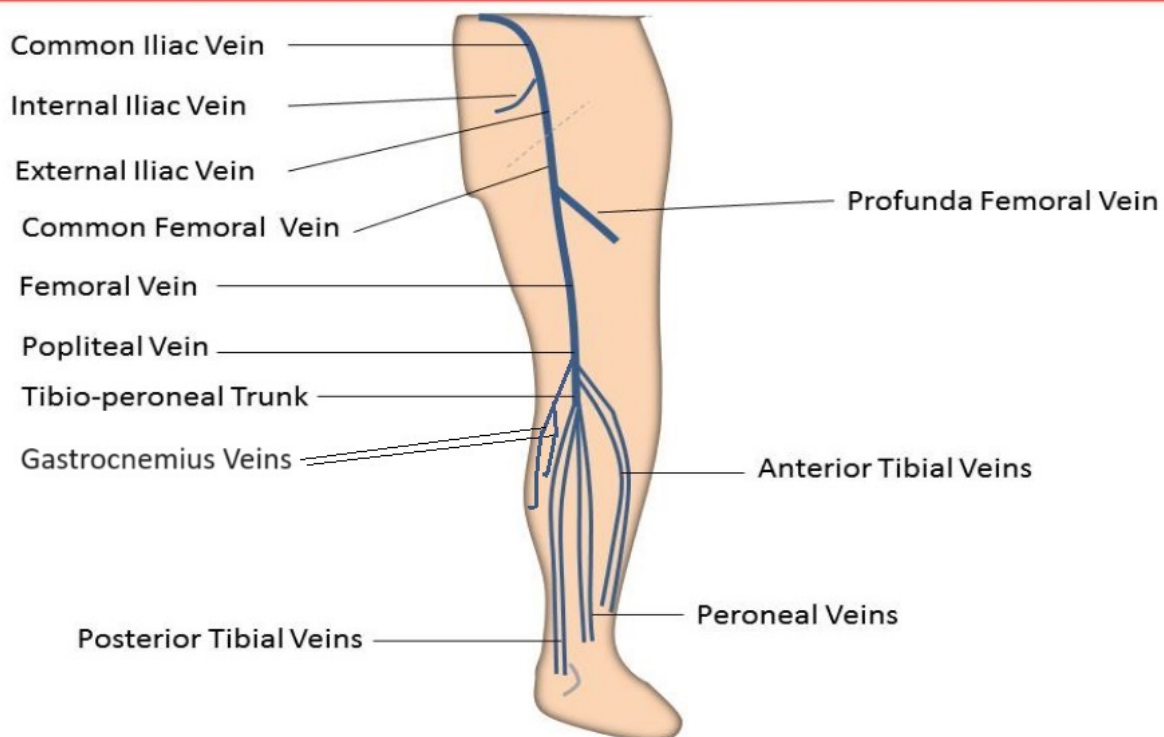
70. It is not possible to evaluate and document valvular insufficiency with only:

- A. Color Doppler Evaluation
- B. Plethysmography
- C. PW Doppler evaluation
- D. 2D real time imaging

71. The gastrocnemius plexus is usually connected to the:

- A. superficial femoral vein
- B. profunda vein
- C. great saphenous vein
- D. popliteal vein

Deep Veins



72. Why would bilateral brachial blood pressure (BP) assessment assist in the diagnosis of a subclavian steal?

- A. The highest brachial pressure is needed to calculate the ABI.
- B. The systolic BP will be lower on the affected side
- C. The systolic BP will be higher in the affected side
- D. To evaluate the potential for future limb amputation.

73. If a renal artery Doppler exam demonstrates stenosis but normal flow is identified on angiography evaluation, the US results are described as:

- A. false positive
- B. true positive
- C. false negative
- D. true negative

74. Bradycardia is defined as a heart rate less than _____bpm.

- A. 45
- B. 50
- C. 60
- D. 65

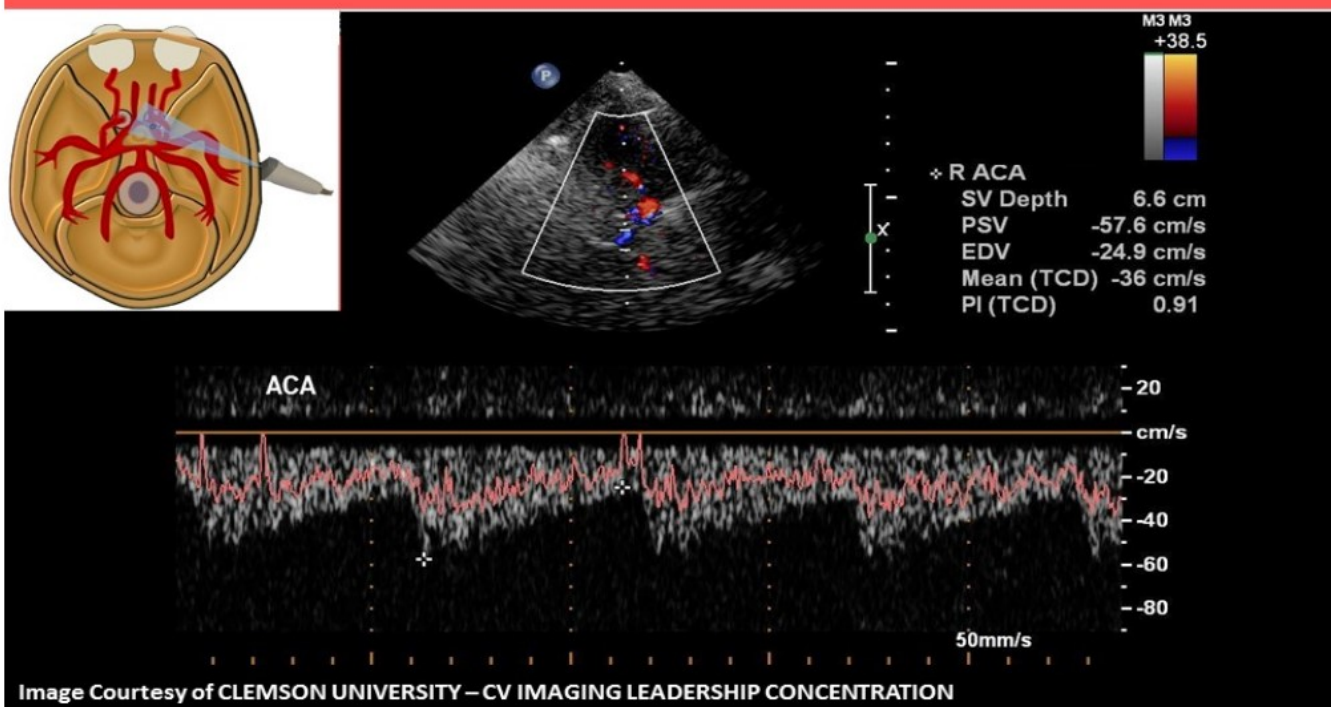
75. What is the gold standard evaluation technique for renal artery stenosis?

- A. MRA
- B. Ultrasound
- C. Angiography
- D. CTA

76. If you detect flow toward the transducer in the right ACA while using the temporal window,

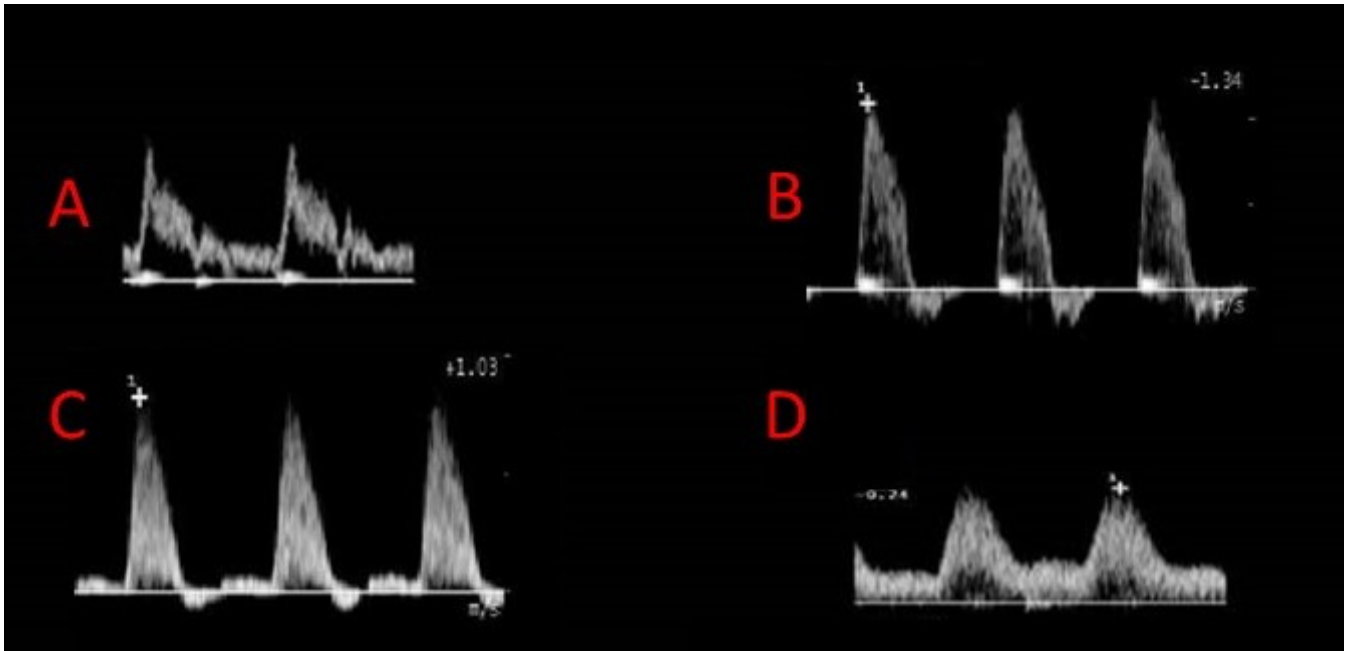
- A. there is probably an occlusion of the right ICA
- B. there is probably a left MCA occlusion
- C. the flow direction is normal
- D. there is probably an occlusion of the left ICA

Transtemporal –ACA



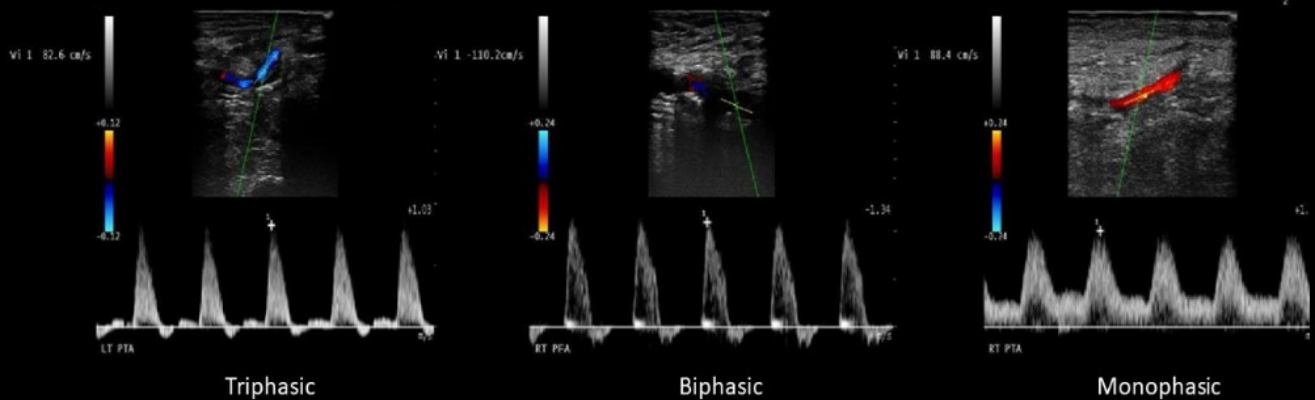
77. Which waveform can be described as biphasic?

- A. A
- B. B
- C. C
- D. D



Arterial Pulsatility

The higher the resistance in the distal tissues, the more pulsatile the flow in the feeding artery. A triphasic Doppler waveform is normally seen in vessels with high resistance (extremities). A monophasic Doppler waveform is normally seen in vessels with low resistance (organs).



78. Venography is usually performed with the patient in what position?

- A. Lithotomy position
- B. Supine with table horizontal
- C. Supine with table at 60-degree Trendelenburg
- D. Supine with table at 60-degree reverse Trendelenburg

79. Arterial claudication occurs:

- A. proximal to the obstruction
- B. while the legs are in the dependent position
- C. while the patient is resting
- D. distal to the obstruction

80. The common femoral artery velocity is 0.8 m/s and the proximal femoral artery demonstrates atherosclerosis formation causing an increase in the velocity to 3.2 m/s. This indicates what level of stenosis?

- A. 75%
- B. 100%
- C. 25%
- D. 50%

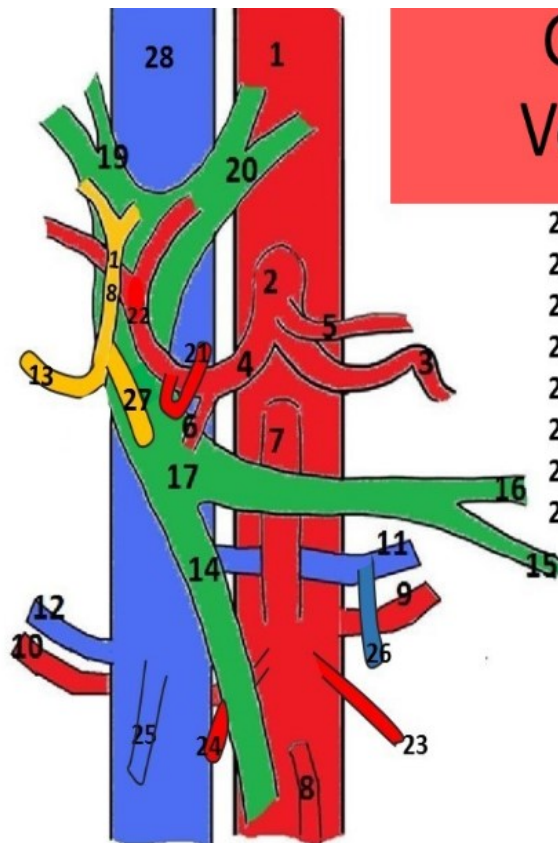
81. If a patient forgets to take their hypertension medication before a renal duplex exam, what effect can this have on the findings?

- A. overestimation of renal parenchymal thickness
- B. underestimation of renal artery stenosis
- C. underestimation of peak renal artery velocity
- D. overestimation of renal artery stenosis

82. Which of the following vessels does not drain into the IVC?

- A. right gonadal vein
- B. right renal vein
- C. left hepatic vein
- D. splenic vein

1. Aorta
2. Celiac
3. Splenic artery
4. Common hepatic artery
5. Left Gastric artery
6. Gastroduodenal artery
7. SMA
8. IMA
9. LRA
10. RRA
11. LRV
12. RRV
13. Cystic duct
14. SMV
15. IMV
16. Splenic Vein
17. Portal confluence
18. Common Hepatic duct
19. RPV
20. LPV



Great Vessels

21. Right Gastric artery
22. Proper hepatic artery
23. Left gonadal artery
24. Right Gonadal artery
25. Right Gonadal vein
26. Left Gonadal vein
27. Common Bile duct
28. IVC

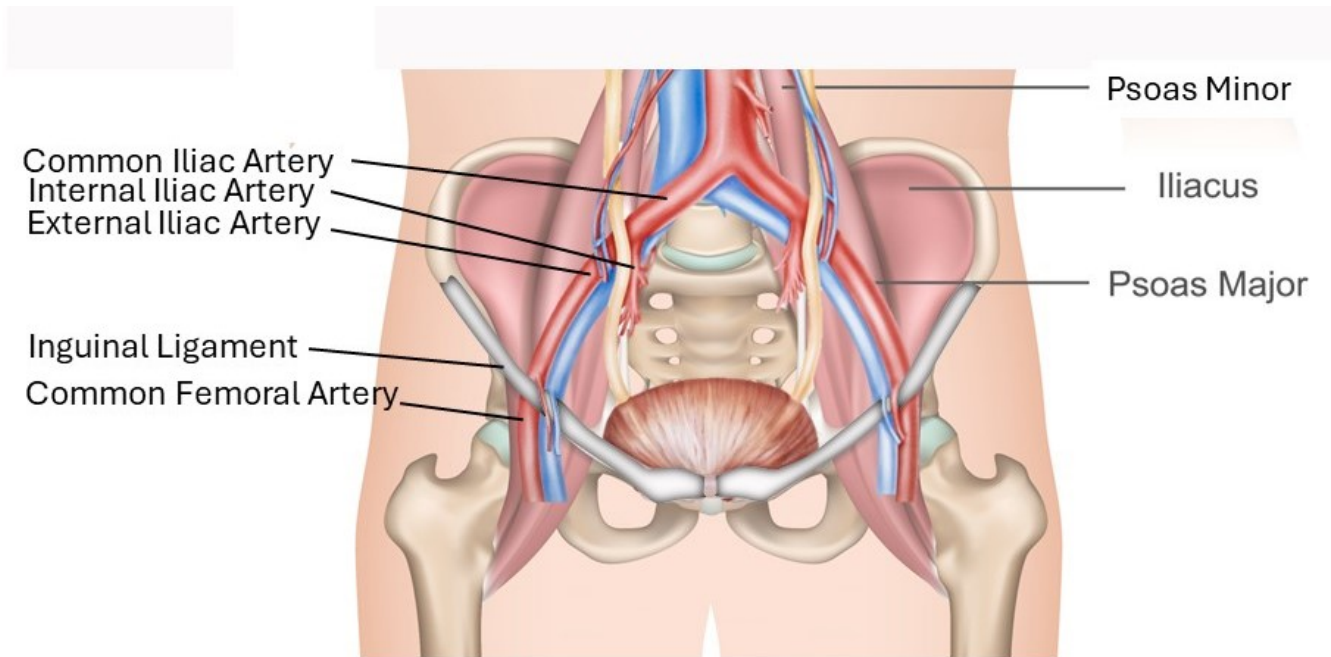
83. A patient presents for a venous insufficiency exam. The exam demonstrates normal bilateral saphenous veins that extend up to the distal thigh, but do not appear to connect to the deep system. What should you review in the patient chart to explain these findings?

- A. medical history for Paget Schroetter syndrome

- B. treatment history for RF ablation of the bilateral GSVs
- C. surgical history for venous ligation of the bilateral GSVs
- D. medical history for Klippel Trenaunay Webber syndrome

84. What Sonographic landmark is used to identify the distal end of the common iliac artery?

- A. the saphenofemoral junction
- B. the bifurcation into the external and internal iliac artery
- C. the inguinal ligament
- D. head of the quadricep muscle



85. A patient presents with a script that requests an aortic Doppler evaluation due to Leriche syndrome. What are you looking for on the exam?

- A. extensive diffuse atherosclerotic disease causing obstruction of the distal aorta
- B. agenesis of the anterior branches of the aorta
- C. monophasic flow in all segments of the aorta due to coarctation of the aorta in the chest
- D. the formation of multiple aneurysms of the aortic segments

86. Which of the following exams is typically performed without angle correction on Doppler evaluation?

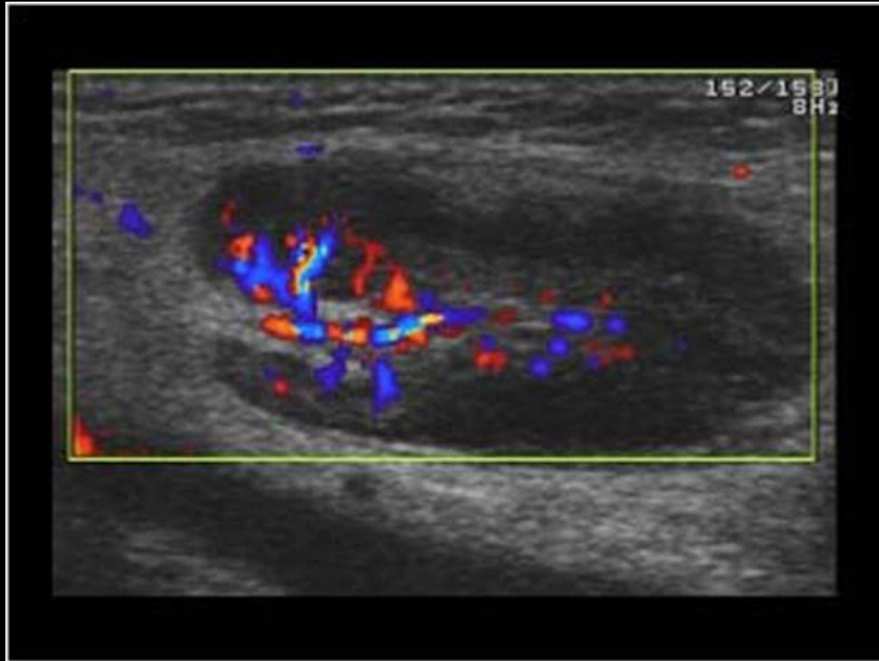
- A. aorta duplex
- B. TIPS evaluation
- C. renal artery duplex
- D. IVC duplex

87. While performing a lower extremity venous exam, you identify a small pair of veins with an accompanying artery deep in the calf muscle, that connect to the popliteal vein. What veins are these?

- A. posterior arch veins
- B. sural veins
- C. anterior tibial veins
- D. veins of Giacomini

88. A patient presents with groin pain 3 days after undergoing right femoral artery angioplasty. The image displayed is from the right groin in the area of pain. What should you report on your technologist worksheet?

- A. A pseudoaneurysm is present that requires immediate intervention
- B. An abnormally enlarged lymph node is present.
- C. A normal lymph node is present.
- D. A pseudoaneurysm is present that is nearly thrombosed



Courtesy of: Dr. Taco Geertsma

www.ultrasoundcases.info

89. Which of the following vessels supply the majority of blood to the cerebellum?

- A. ICA and Vertebral
- B. ICA and ECA
- C. Vertebral and Basilar
- D. ECA and Basilar

90. Which of the following will maintain compliance with the ALARA principle?

- A. reducing overall gain
- B. reducing the number of images taken in standard protocol
- C. reducing transducer frequency
- D. reducing the amount of protected information given to the patient

91. There are normally _____ perforator veins in each leg.

- A. about 25-30
- B. about 100
- C. about 300
- D. over 500

92. Which of the following correctly describes a pre- procedure vein mapping?

- A. The diameter of the vein is usually written on the arm/leg at the location it was taken

- B. Veins should maintain a uniform elliptical shape during measurements
- C. Keep the transducer pressure constant at a medium level
- D. Larger amounts of gel will aid in visualization of the superficial vessels

93. The Achilles tendon is a landmark for which deep calf vein?

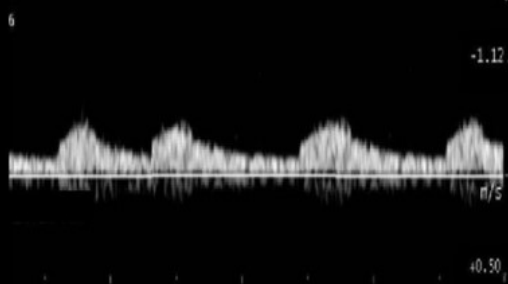
- A. anterior tibial vein
- B. small saphenous vein
- C. posterior tibial vein
- D. peroneal vein

94. Which of the following can lead to a false negative diagnosis of ICA stenosis?

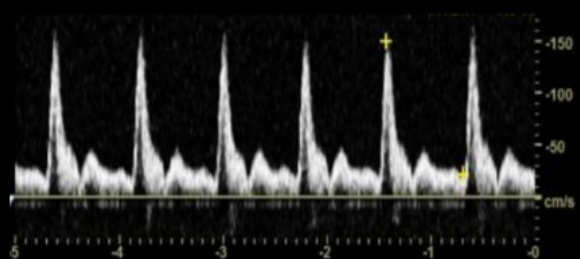
- A. Congestive heart failure
- B. Increased cardiac output
- C. Tortuosity
- D. Anemia

Cardiac Output

Coronary artery disease, dilated cardiomyopathy, and severe mitral stenosis are common causes of low cardiac output. Late peaking, lower velocity, lower resistance flow will be displayed in the carotid arteries.

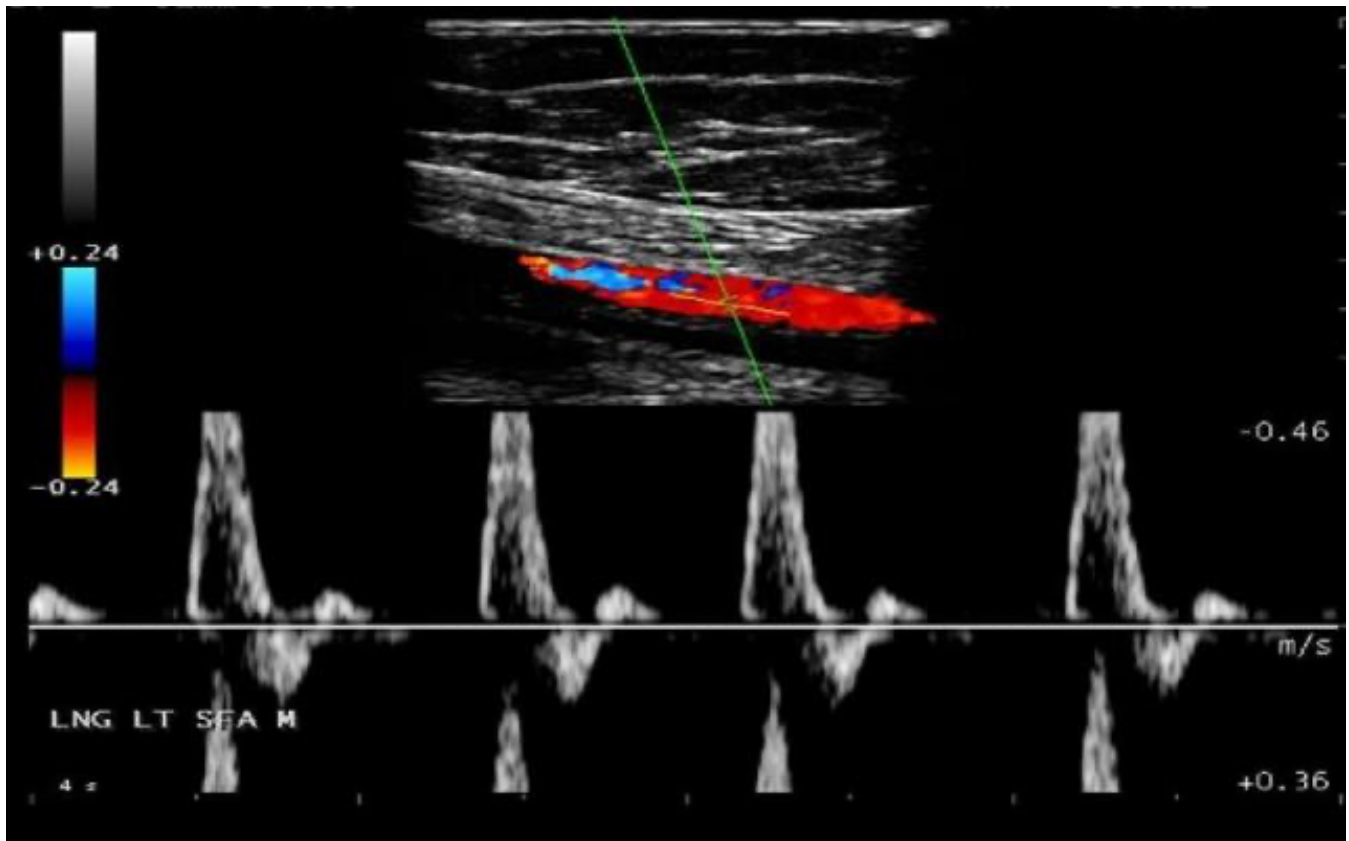


Anemia, hyperthyroidism, and pregnancy are common causes of high cardiac output. Sharp peaking, higher velocity, higher resistance flow will be displayed in the carotid arteries.



95. What adjustments can be made to improve the attached image of the Doppler tracing to make it of diagnostic quality?

- A. Increase the sample volume size
- B. Decrease velocity scale
- C. Lower the baseline
- D. Increase the sweep speed

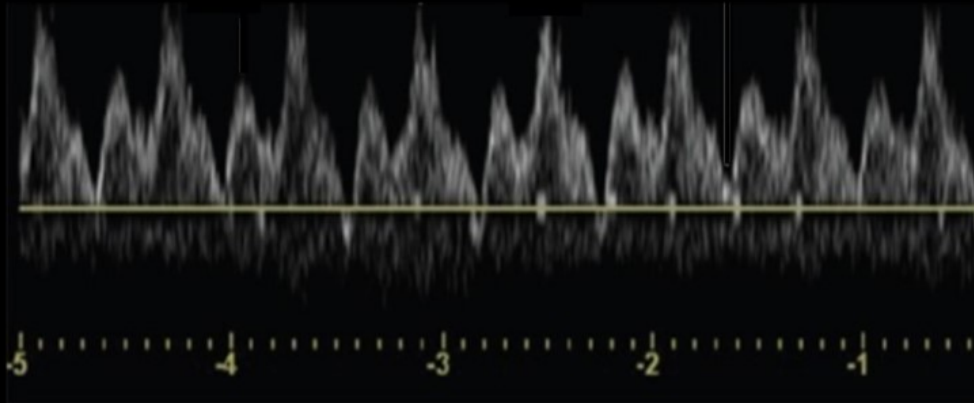


96. A patient presents for a carotid ultrasound. The chart states there is a history of dilated cardiomyopathy and an intra-aortic balloon pump is present. Which of the following correctly describes how to evaluate the patient for carotid stenosis?

- A. The patient should be rescheduled until after the pump is removed because there is no way to evaluate carotid stenosis while a pump is active
- B. The peak systolic velocity is invalid, but the end diastolic velocity should be measured
- C. Waveforms from the common carotid artery will have two systolic peaks and the first one should be measured to determine the peak velocity
- D. Waveforms from the common carotid artery will have two systolic peaks and the second one should be measured to determine the peak velocity

Intra-aortic Balloon Pump

Carotid Waveform



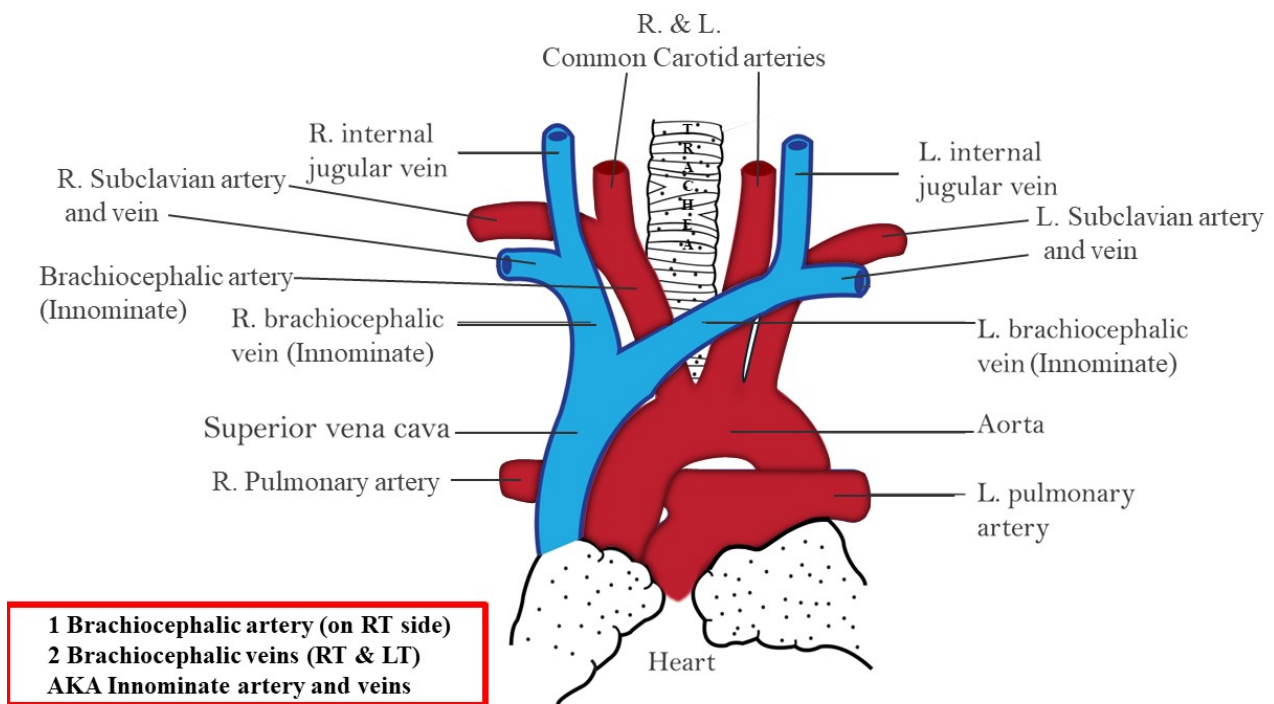
Adapted from: Ginat, D., Massey, H. T., Bhatt, S., & Dogra, V. S. (2011). Imaging of mechanical cardiac assist devices. *Journal of clinical imaging science*, 1, 21. doi:10.4103/2156-7514.80373 Copyright-open license

97. Which of the following conditions would be treated with streptokinase?

- A. Phlegmasia cerulea dolens
- B. 70-90% carotid stenosis
- C. Venous insufficiency
- D. Partial DVT

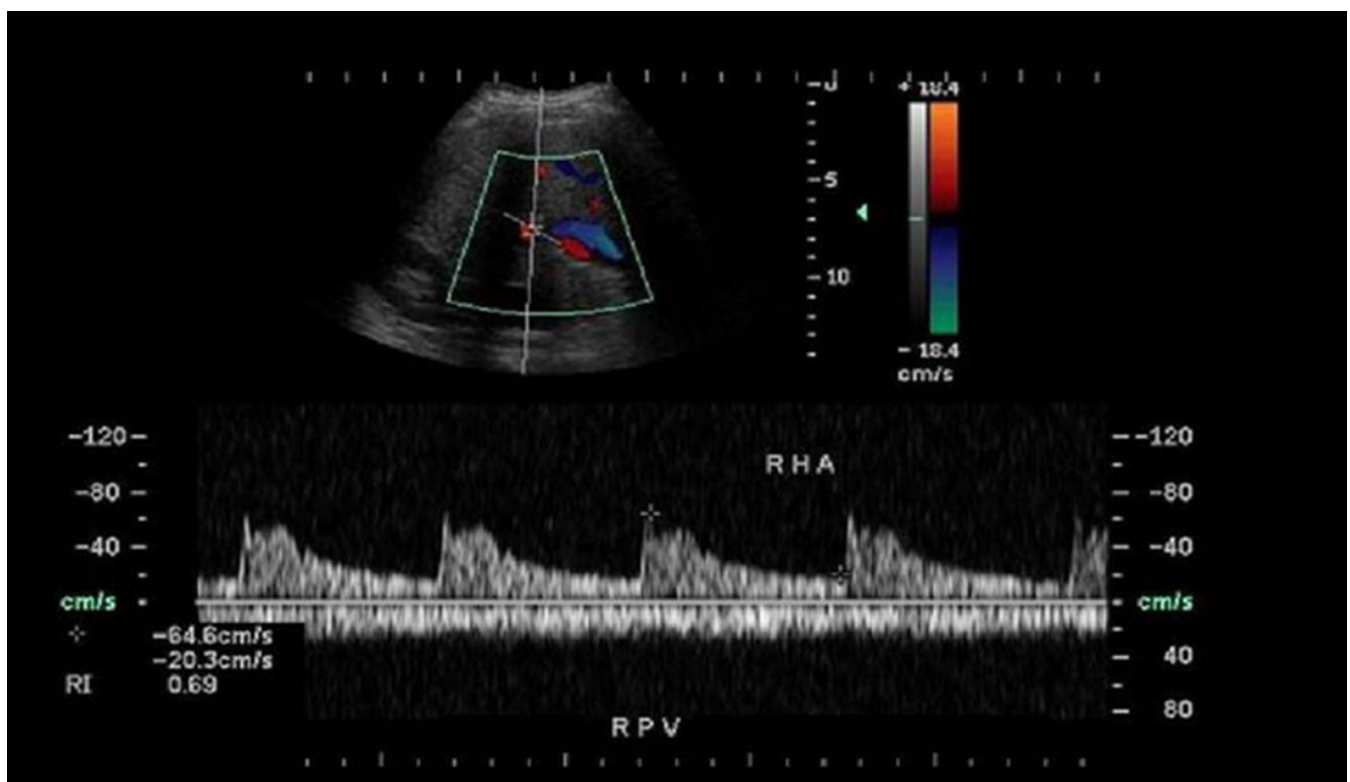
98. The innominate vein is formed by the junction of:

- A. superior vena cava and inferior vena cava
- B. internal jugular vein and superior vena cava
- C. subclavian vein and cephalic vein
- D. subclavian vein and internal jugular vein



99. Which of the following statements is true regarding the Doppler evaluation of the liver vasculature displayed?

- A. The hepatic artery and portal vein are demonstrated with normal flow directions.
- B. The hepatic artery is hepatopetal and the portal vein is hepatofugal.
- C. The hepatic artery is hepatofugal and the portal vein is hepatopetal.
- D. The portal vein flow is normal, but the hepatic artery demonstrates increased resistance



100. Which of the following correctly describes cuff placement for a segmental pressure exam?

- A. The width of the cuff should be at least 20% narrower than the girth of the limb segment
- B. Cuffs should be placed straight around the limb, rather than angled to the curve of the limb
- C. Once the patient is supine, ask him/her to raise their foot about 6 inches above the table to you can properly wrap the cuff around the leg
- D. The proximal edge of the cuff should always cover a bony structure like the malleolus or tibial tubercle

101. The best obtainable angle of insonation for vascular evaluation that is most commonly used to provide Doppler shift information is _____ degrees.

- A. 0
- B. 45
- C. 60
- D. 90

102. Which of the following describes a cardiovascular application of the Bernoulli Principle?

- A. explains the relationship of vessel radius and flow volume
- B. explains the relationship between pressure, resistance and flow volume
- C. explains the turbulence and layers of flow separation within the carotid bulb
- D. explains the reason for aliasing of the Doppler signal at high velocities

103. Which of the following correctly describes compartment syndrome?

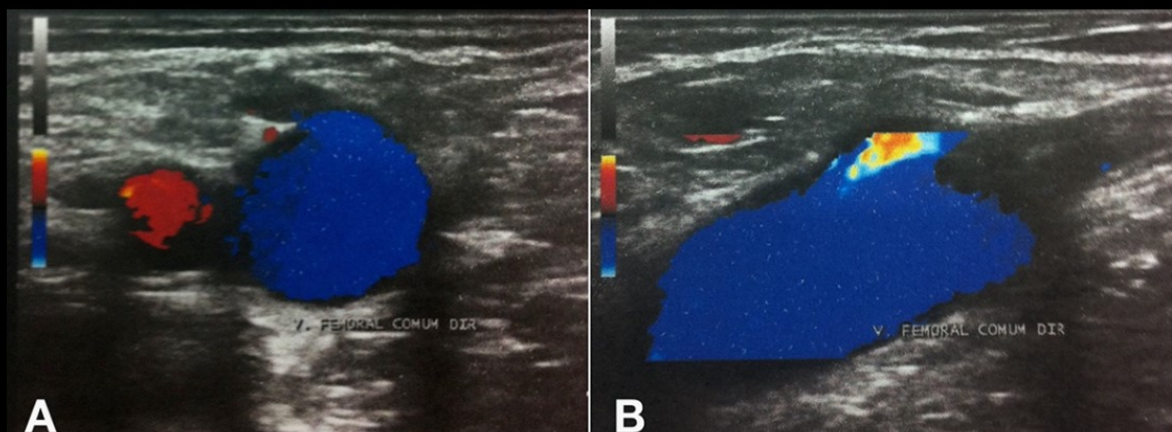
- A. most commonly occurs due to trauma
- B. typically occurs below the diaphragm in the chest
- C. caused by compression of nerves during normal extremity motion
- D. typically occurs below the diaphragm in the abdomen

104. Endovenous heat-induced thrombosis is a complication of:

- A. chemical ablation
- B. microphlebectomy
- C. sclerotherapy
- D. radiofrequency ablation

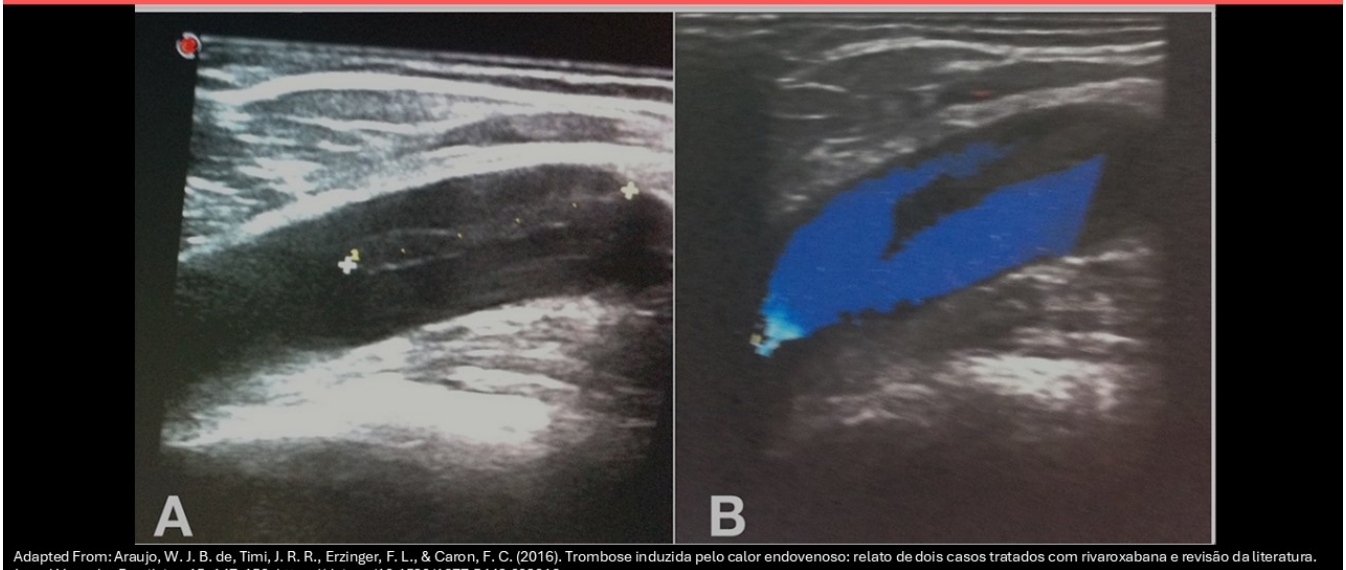
Endovenous Heat Induced Thrombus (EHIT)

If the catheter is operated too close to the SFJ, thrombus can form and extend into the deep venous system



Adapted From: Araujo, W. J. B. de, Timi, J. R. R., Erzinger, F. L., & Caron, F. C. (2016). Trombose induzida pelo calor endovenoso: relato de dois casos tratados com rivaroxabana e revisão da literatura. *Journal Vascular Research*, 15, 147-152. <https://doi.org/10.35904/1677-5449.000016>

Endovenous Heat Induced Thrombus (EHIT)



105. Which of the following is the second most common site for atherosclerosis formation in the lower extremity?

- A. distal SFA in Adductor canal
- B. popliteal trifurcation
- C. common carotid artery bifurcation
- D. common femoral artery bifurcation

106. Bilateral bruits are detected in the proximal neck. This finding is most suggestive of:

- A. Aortic valve stenosis
- B. Subclavian steal
- C. Polycythemia vera
- D. CHF

107. In most patients with a central venous catheter the tip of the catheter is positioned in the:

- A. proximal subclavian vein
- B. internal jugular vein
- C. distal subclavian vein
- D. distal 1/3 of the superior vena cava

108. The parenchymal resistance ratio is calculated to assess perfusion of which of the following organs?

- A. brain
- B. kidney
- C. muscles
- D. liver

109. Which vein is best evaluated for compressibility by asking the patient to perform a quick, deep sniff?

- A. subclavian vein

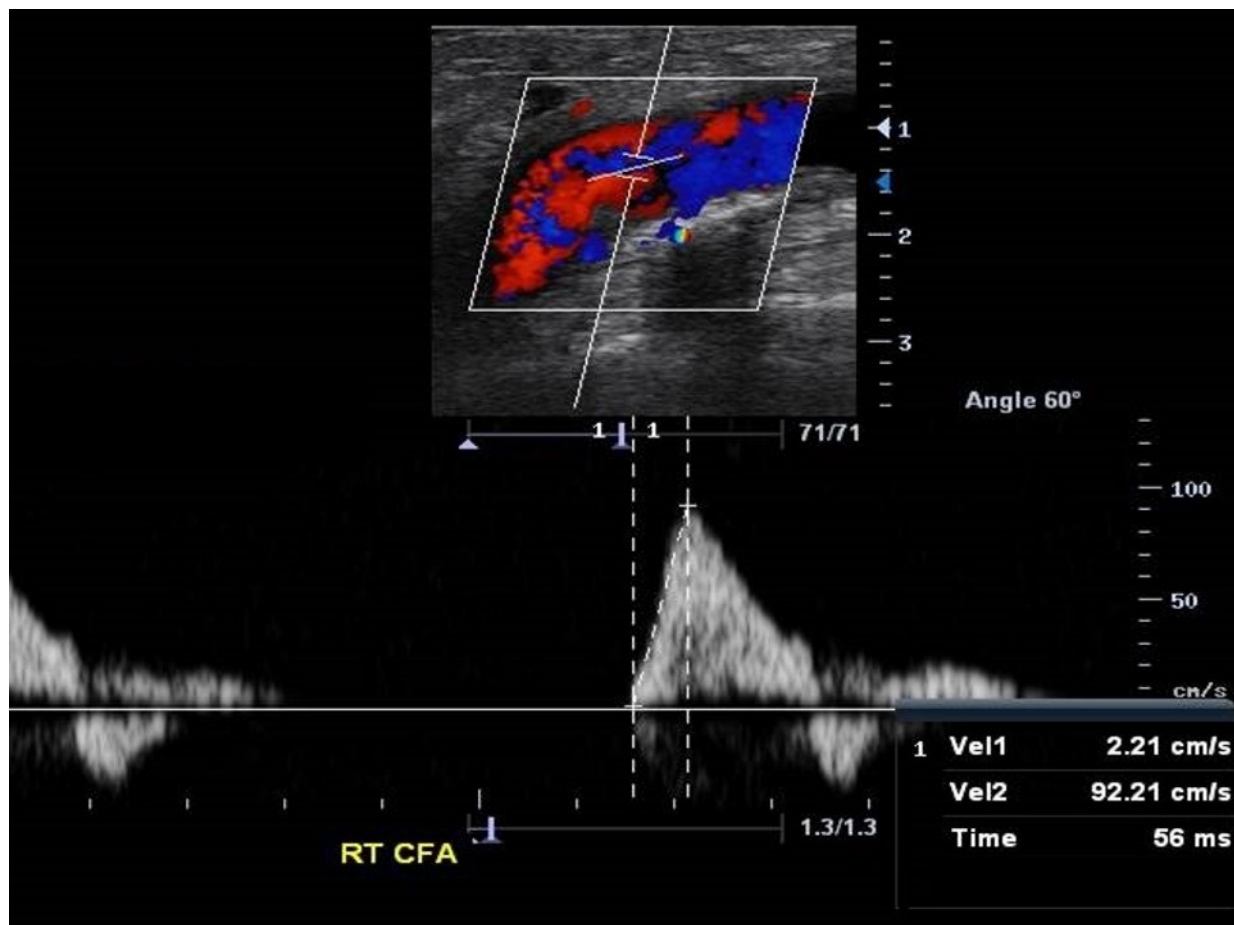
- B. great saphenous vein
- C. distal IVC
- D. axillary vein

110. Which of the following lower extremity vessels would most likely be occluded in a patient with Phlegmasia Alba Dolens?

- A. deep veins, superficial veins and common femoral artery
- B. saphenous veins only
- C. common femoral artery
- D. femoral vein, deep calf veins, popliteal vein

111. What blood flow characteristic is being measured on the Doppler tracings provided?

- A. Resistive index
- B. Pulsatility index
- C. Acceleration time
- D. Peak systolic velocity



112. The brightness of the reflection on the Doppler tracing represents _____.

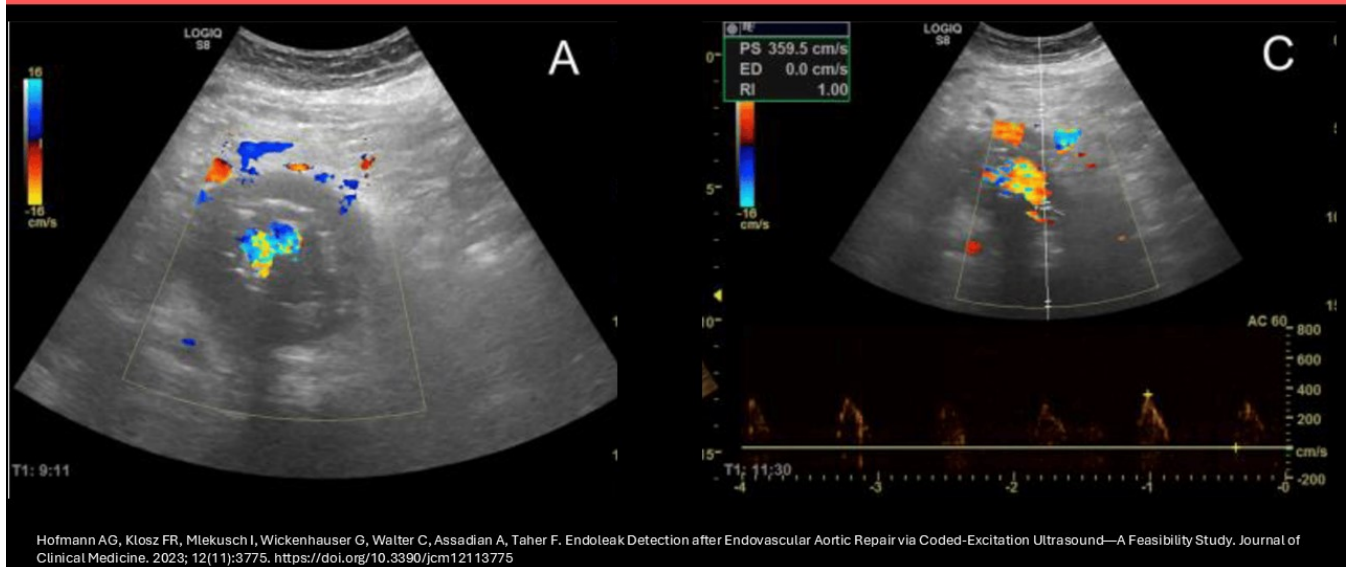
- A. the time duration of the reflection
- B. the spatial pulse length of the reflection
- C. the number of red blood cells moving at each velocity displayed
- D. the frequency of the reflection

113. Color Doppler evaluation of an aortic endograft:

- A. is the most accurate method for diagnosis of an endoleak

- B. should be performed with increased wall filter and velocity scale settings with reduced color priority
- C. cannot be performed on patients who are less than 3 months post-op due to the risk of cavitation and graft migration
- D. can be limited by twinkle artifact and color flash artifact

EVAR – Artifacts on Color Doppler

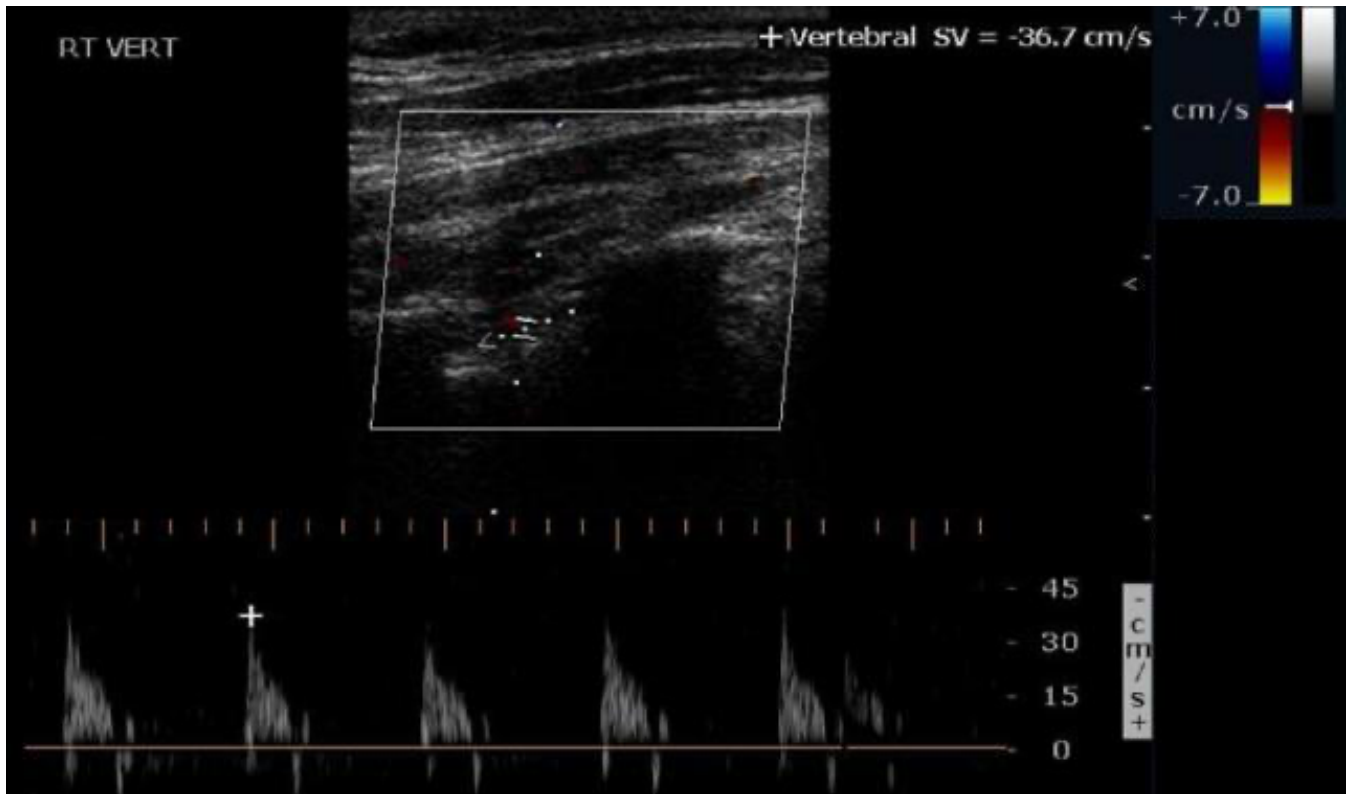


114. The great saphenous vein originates _____

- A. at the dorsum of the foot and courses anterior to the lateral malleolus.
- B. at the dorsum of the foot and courses anterior to the medial malleolus.
- C. at the great toe and courses along the ventral aspect of the foot to reach the medial ankle and extend up the leg.
- D. at the great toe and courses along the ventral aspect of the foot to reach the lateral ankle and extend up the leg.

115. The image demonstrated is from a routine carotid ultrasound. Based on this image, what should be done next to evaluate this patient?

- A. Evaluate the right subclavian artery for the presence of monophasic flow
- B. Perform a TCD exam to look for a basilar obstruction
- C. Evaluate the left subclavian artery for the presence of monophasic flow
- D. Perform a TCD exam to look for a contralateral MCA for obstruction

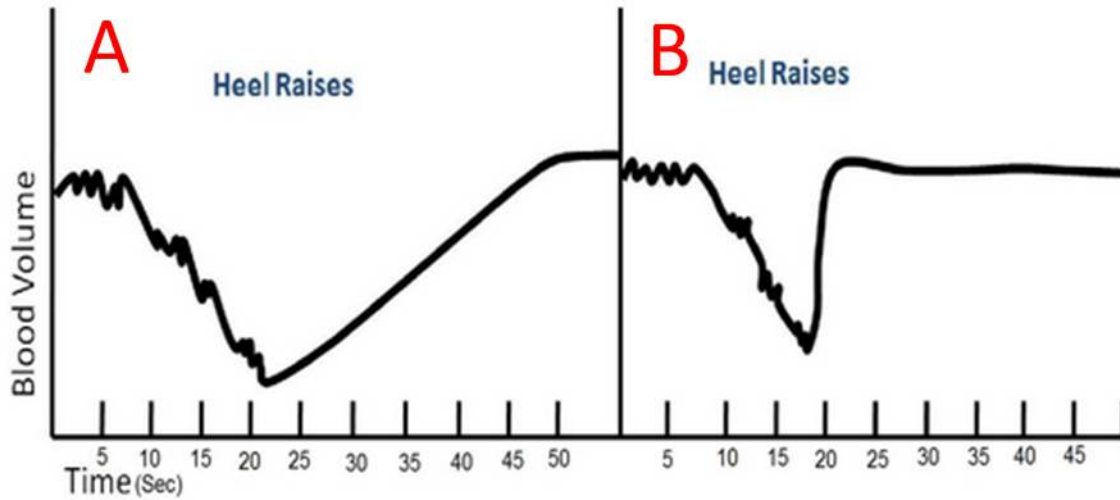


116. Which of the following is supplied with arterial blood through branches of the ECA?

- A. cerebellum
- B. eyes
- C. thyroid gland
- D. brain stem

117. Which drawing demonstrates an abnormal result for a venous photoplethysmography exam?

- A. A
- B. B
- C. Both are abnormal responses
- D. Unable to determine without the application of a tourniquet on the thigh



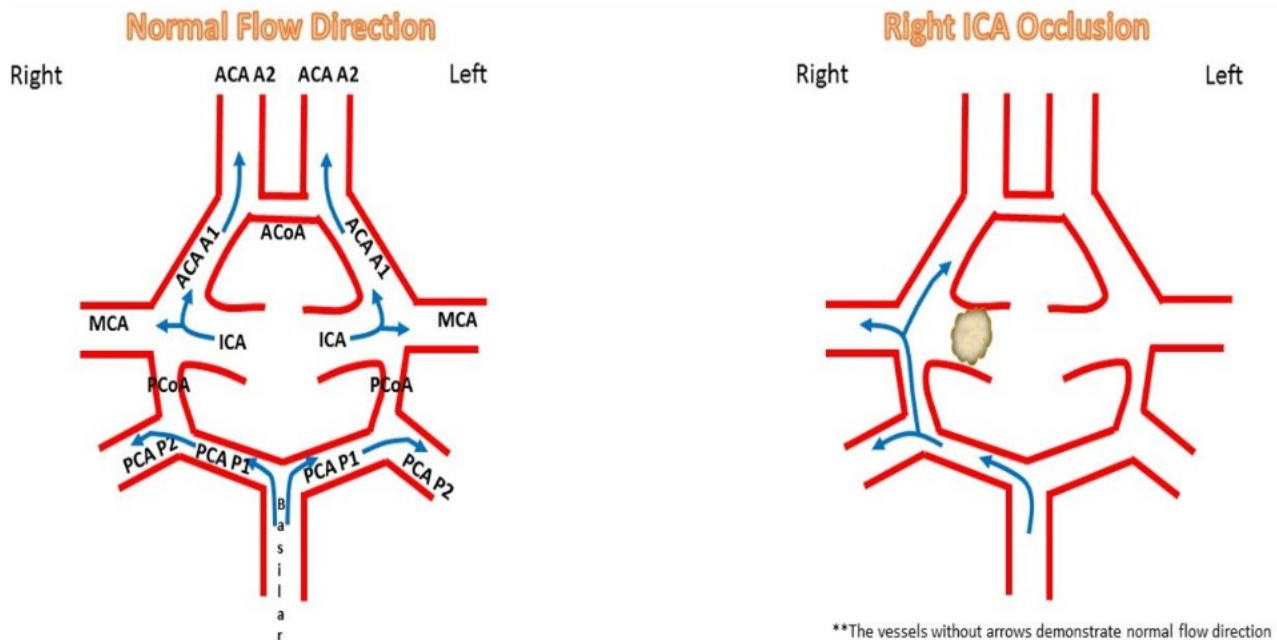
118. Hydrostatic venous pressure _____ when a patient goes from standing to the supine position.

- A. shows a quick decrease before returning to 100mmHg
- B. increases
- C. decreases
- D. will show little change

119. Which vessel can provide a pathway for collateral flow from the vertebral system to the MCA on the same side of the body?

- A. anterior cerebral artery
- B. posterior communicator arteries
- C. basilar artery
- D. anterior communicator arteries

Posterior to Anterior Collateralization



120. The Doppler tracing of the flow in a vessel just distal to a stenosis will demonstrate _____ than at the stenosis.

- A. greater resistance
- B. greater bandwidth
- C. more laminar flow
- D. greater Doppler shift

121. A 65yr old female presents with left leg swelling for over one year. Mild hyperpigmentation is present in the distal calf and ankle area. These clinical findings are most suggestive of:

- A. Cellulitis
- B. Superficial Phlebitis
- C. Chronic Insufficiency
- D. Acute DVT

122. The most common color change demonstrated in a lower extremity with an acute arterial occlusion is:

- A. cyanosis
- B. pallor
- C. rubor
- D. brawny

123. Bank tellers, teachers, cashiers and assembly line workers have an increased risk of all of the following, except:

- A. varicose veins

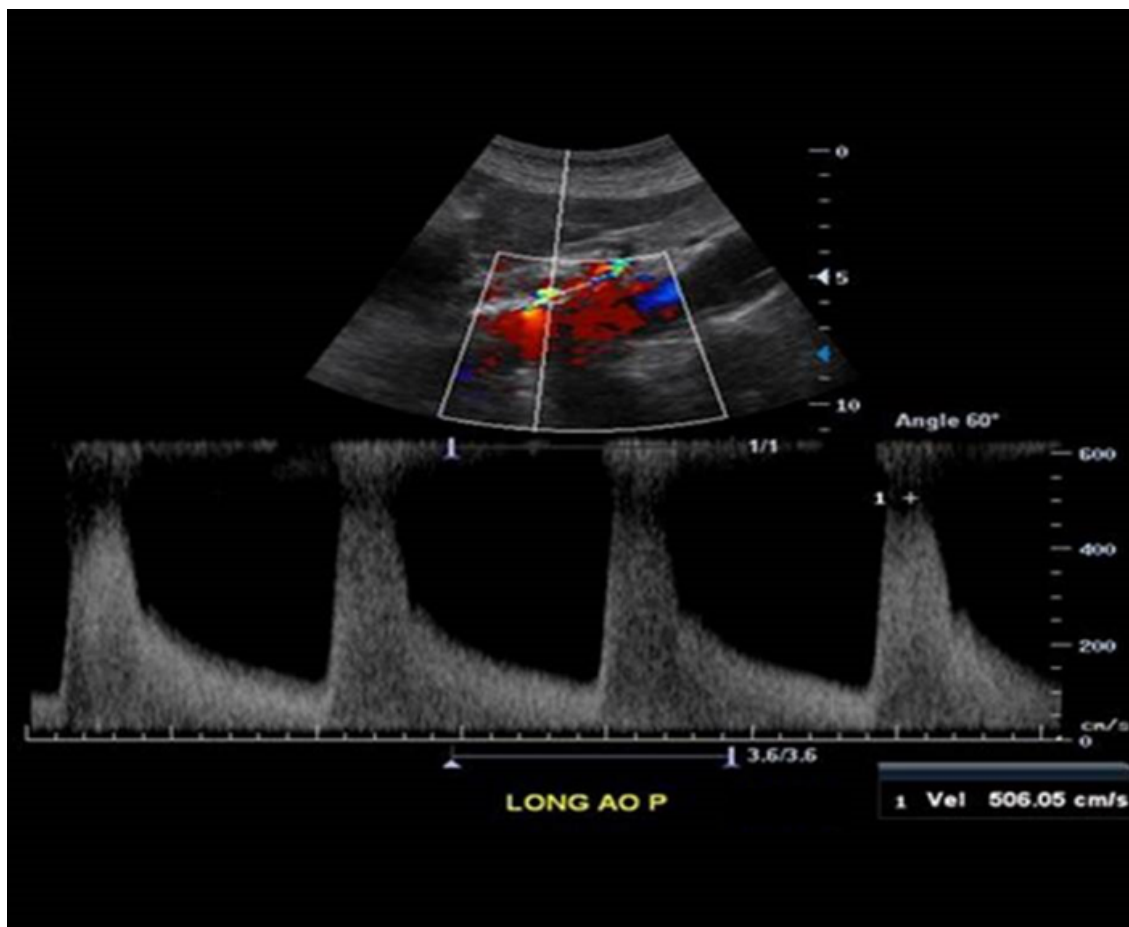
- B. jugular vein thrombosis
- C. thrombus formation in the soleal sinuses

124. An end diastolic velocity of 160 cm/s is an indication of _____ in the ICA.

- A. normal flow
- B. mild stenosis
- C. moderate stenosis
- D. severe stenosis

125. This Doppler tracing taken from a branch of the aorta demonstrates:

- A. Critical stenosis of the celiac axis
- B. Hemodynamically significant stenosis of the right renal artery
- C. Normal celiac flow with top normal PSV
- D. Hemodynamically significant stenosis of the left renal artery



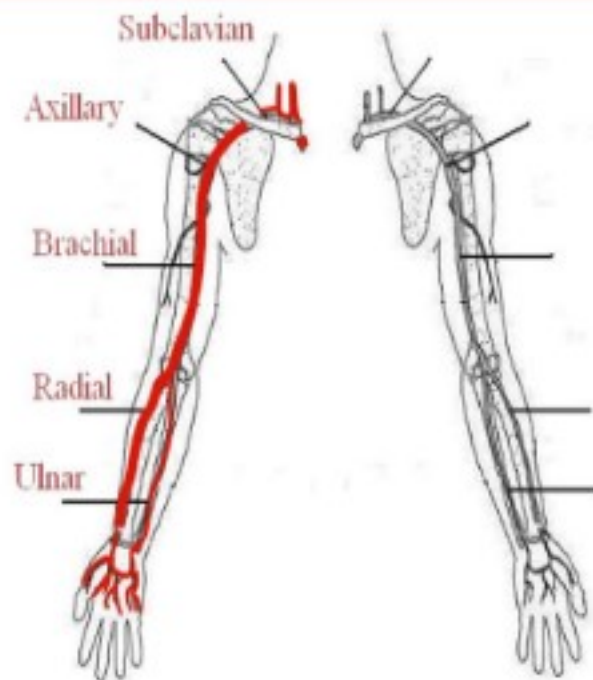
126. The type of plaque that carries the greatest risk of embolization is _____.

- A. soft
- B. complex
- C. calcific
- D. soft and complex

127. Which arteries supply the forearm and hand with blood?

- A. 2 Radial and 2 Ulnar Arteries
- B. 1 Radial Artery and 1 Ulnar Artery
- C. 1 Cephalic Artery and 1 Basilic Artery

Upper Extremity Anatomy



128. When evaluating suspected thoracic outlet syndrome, a PPG exam can be performed with the sensor placed on the _____ or a Doppler exam can be performed with the transducer placed _____.

- A. 2nd digit, on the radial artery
- B. lateral wrist, on the brachial artery
- C. medial wrist, on the radial artery
- D. 1st digit, on the carotid artery

129. A segmental pressure exam is performed with the following results. Lt arm 140mmHg Lt ankle 155mmHg Rt arm 110mmHg Rt ankle 145mmHg Which of the following correctly describes the findings?

- A. There is no vascular disease present in the right leg, but the left leg demonstrates mild disease.
- B. There is mild vascular disease present in the lower extremities.
- C. There is mild vascular disease present in the lower extremities and there is most likely a subclavian steal present.
- D. There is no vascular disease present in the legs, but there is most likely a subclavian steal present.

130. Which of the following describes the expected change in the PVR tracing on an immediate post procedure evaluation of a patient after successful revascularization of the common femoral artery?

- A. increased amplitude by >100%
- B. increased amplitude by >50%
- C. decreased amplitude by >50%
- D. loss of the dicrotic notch

131. Which of the following is most suggestive of a subclavian steal?

- A. amaurosis fugax
- B. systemic HTN
- C. difference in brachial pressures >20 mmHg
- D. congestive heart failure

132. Which of the following sonographic characteristics is most suggestive of renal vein thrombosis?

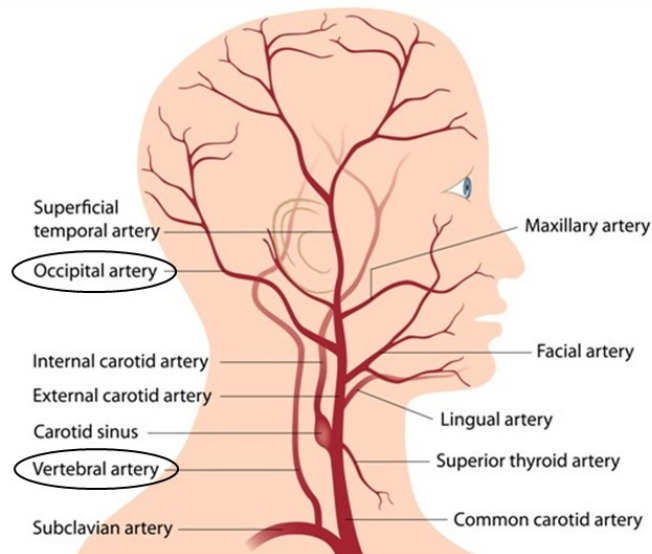
- A. reduced diastolic flow in the renal arteries
- B. prominent flow in the segmental and intralobar arteries
- C. turbulent venous flow within the kidney
- D. low resistance flow in the intrarenal arteries

133. Branches of the _____ connect to branches of the vertebral arteries to allow collateral flow from the anterior circulation to the posterior circulation.

- A. occipital artery
- B. basilar artery
- C. ophthalmic artery
- D. subclavian artery

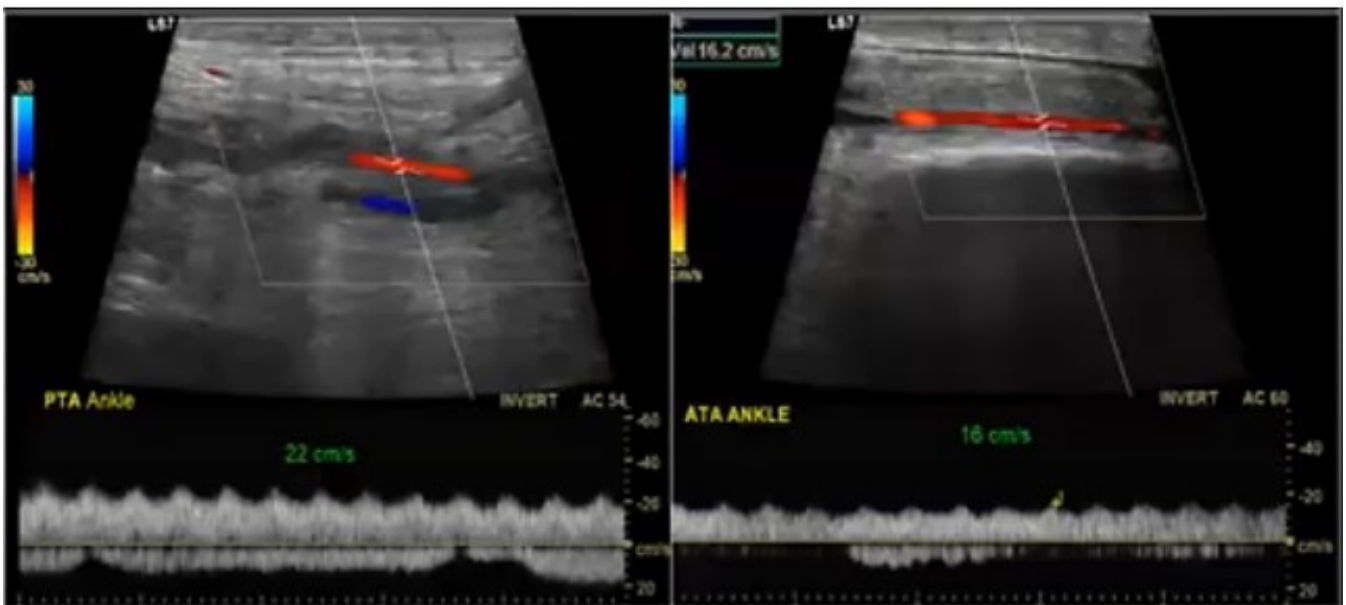
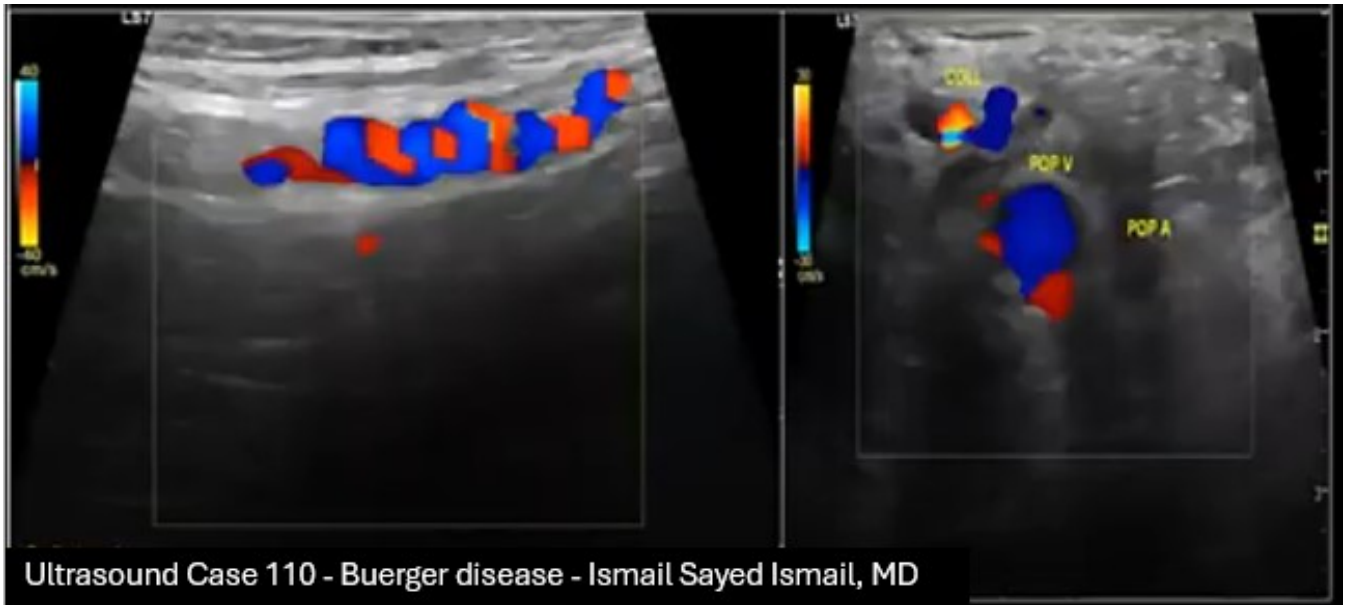
Head and Neck Arteries

Small branches of the occipital artery connect to branches of the vertebral artery. This can serve as a collateral pathway from the ECA (anterior circulation) to the vertebral artery posterior circulation) or vice versa.



134. You receive an order for an upper extremity arterial exam to rule out Buerger disease. What is the preferred method of evaluation?

- A. Evaluate the hands for vascular changes with cold sensitivity testing
- B. Evaluate the arterial flow at rest and with exercise
- C. Evaluate the smaller arteries of the arm with Duplex and PPG techniques
- D. Evaluate the arteries of the shoulder for extrinsic compression with movement



135. The external carotid artery normally demonstrates a velocity below:

- A. 100cm/s
- B. 80cm/s
- C. 150cm/s
- D. 230cm/s

136. A patient presents for a lower extremity venous Doppler to rule out venous insufficiency. Which vessels should be completely evaluated for the exam?

- A. iliac, femoral and great saphenous veins
- B. femoral and popliteal veins
- C. femoral, popliteal, short saphenous and great saphenous veins
- D. great and short saphenous veins

137. The image is a transverse view of right shoulder. What vessel is indicated by letter B?

- A. cephalic vein
- B. basilic vein
- C. axillary vein

D. subclavian vein



138. Venous ulcers are normally located on the _____, while arterial ulcers are normally found on the _____.

- A. calf, thigh
- B. heel, toes
- C. medial calf/ankle, toes and heel
- D. toes and heel, medial calf/ankle

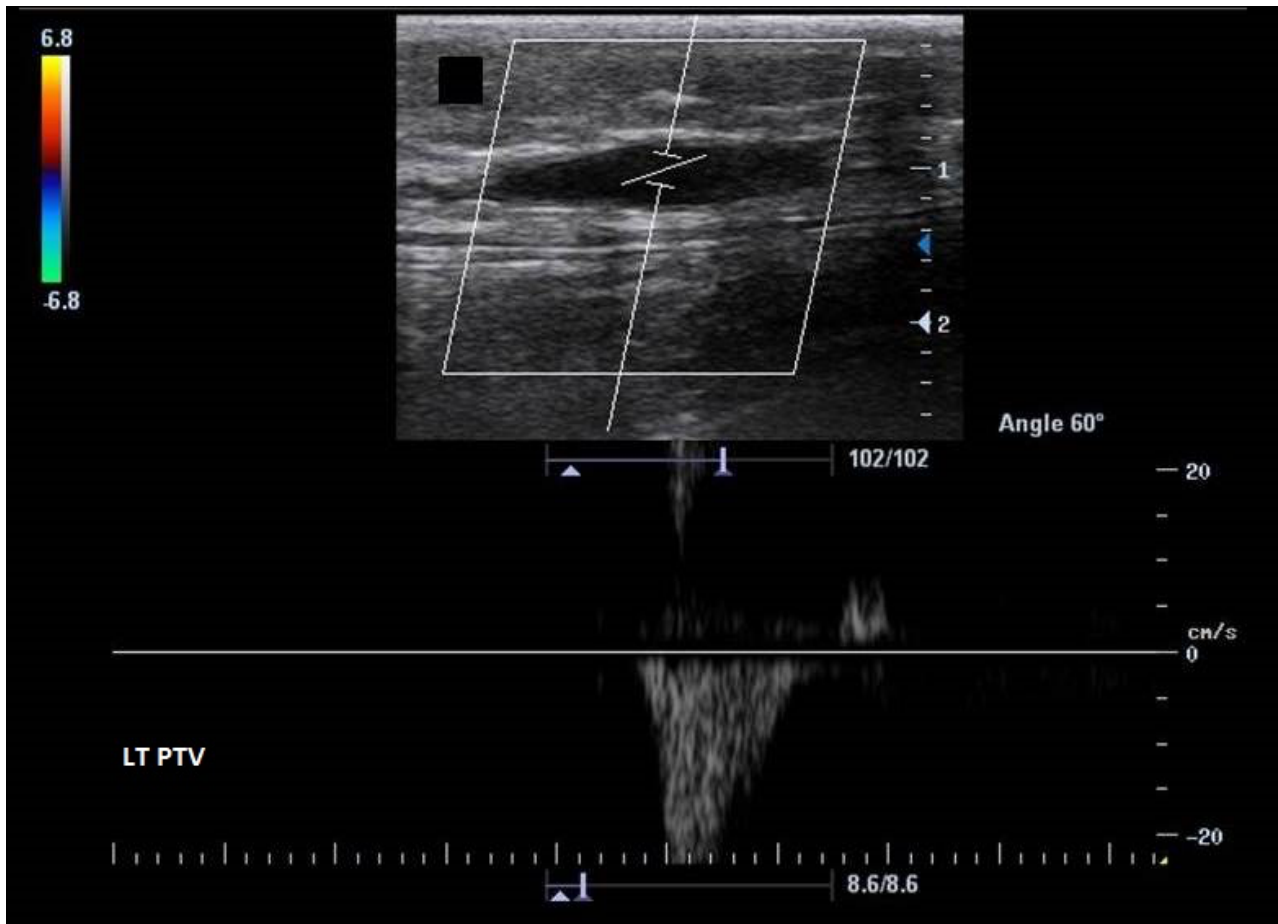
Venous Ulcers	Arterial Ulcers
Mild pain	Severe pain
Shallow ulceration	Deep ulceration
Wet and oozing	Dry, scab formation
Typically found near medial malleolus	Typically found near lateral malleolus, heel and toes

139. A 65yr old male presents with mild left calf pain after walking 10 blocks. The pain is relieved by sitting and resting. On a resting arterial exam with ABI, the left ABI is 1.04. These findings are most suggestive of:

- A. An exercise arterial should be performed to re-evaluate the ABI of the left leg before a diagnosis is made.
- B. The venous system should be evaluated for possible insufficiency.
- C. The venous system should be evaluated for thrombosis.
- D. The patient is most likely experiencing an orthopedic or neurogenic problem and an MRI may be warranted.

140. The attached image is taken from a lower extremity venous Doppler exam. Which of the following should be included in your preliminary report?

- A. The left PTV demonstrates abnormal absence of spontaneous flow but normal response to proximal augmentation.
- B. The left PTV demonstrates normal flow and response to distal augmentation.
- C. The left PTV demonstrates normal flow and response to proximal augmentation.
- D. The left PTV demonstrates abnormal absence of spontaneous flow but normal response to distal augmentation.



141. When a patient has multiple indications for an ultrasound exam, which one is used to determine appropriate use of vascular ultrasound testing?

- A. the indication with the lowest score should be used to determine appropriate use
- B. if a patient has multiple symptoms, the Appropriate Use Criteria is invalid
- C. all scores should be added together and divided by the number of indications
- D. the indication with the highest score should be used to determine appropriate use

142. Which of the following arteries will be compressed when evaluating frontal artery flow for a periorbital Doppler exam?

- A. external carotid artery and vertebral artery
- B. superficial temporal artery and infraorbital artery
- C. ophthalmic artery and contralateral common carotid artery
- D. internal carotid artery and vertebral artery

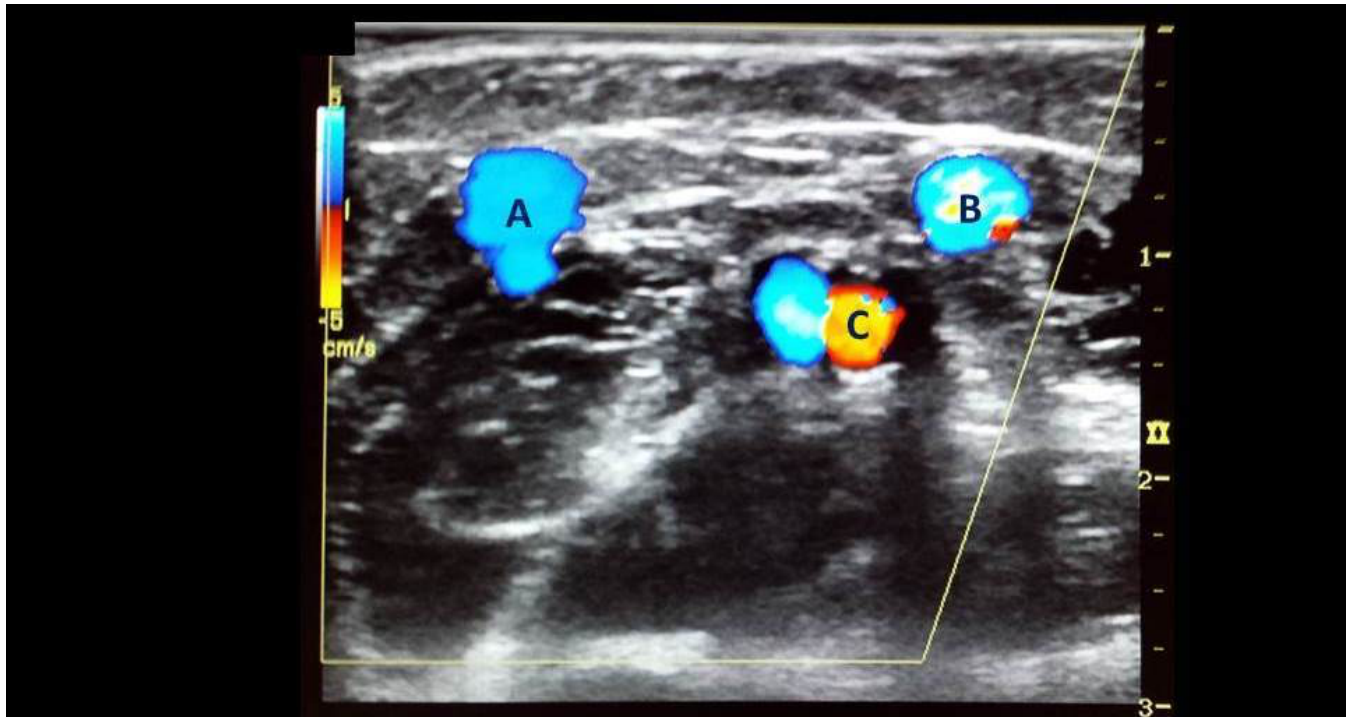
143. A normal penile brachial index is:

- A. >0.8
- B. >0.5
- C. >0.75
- D. >0.6

144. The image is a transverse view of the right antecubital fossa. What vessel is indicated by letter A?

- A. basilic vein
- B. cephalic vein
- C. median cubital vein

D. brachial vein



145. Which of the following is considered a limitation of the three cuff lower extremity segmental pressure exam?

- A. There is no way to differentiate proximal and distal calf artery disease.
- B. The single thigh cuff normally overestimates the thigh pressure.
- C. There is no way to differentiate lower thigh and popliteal arterial disease.
- D. There is no way to differentiate groin and ankle arterial disease.

146. An ABI of 0.77 indicates _____ disease.

- A. severe
- B. unable to determine without exercise portion
- C. minimal
- D. mild - moderate

147. Lower extremity arterial segmental pressure exam results in a post exercise basal pressure recovery time of 2 to 6 minutes which suggests:

- A. Thoracic outlet syndrome
- B. Single level stenosis
- C. Iliac Disease
- D. Multi-level stenosis

148. Which of the following is an expected finding with an AV fistula caused by an interventional procedure?

- A. continuous venous outflow
- B. direct connection between a single artery and vein
- C. dilated venous collaterals adjacent to the fistula
- D. high resistance arterial inflow

149. A patient presents with HTN, DM and left lower extremity pain that worsens as he walks. He usually begins to have pain after he walks to his mailbox with increasing pain as he

returns to the house. The pain is relieved by resting his leg for a few minutes. A bilateral resting ABI is performed with the right leg, 1.12 and the left leg 1.34. Which of the following statements is true regarding the findings?

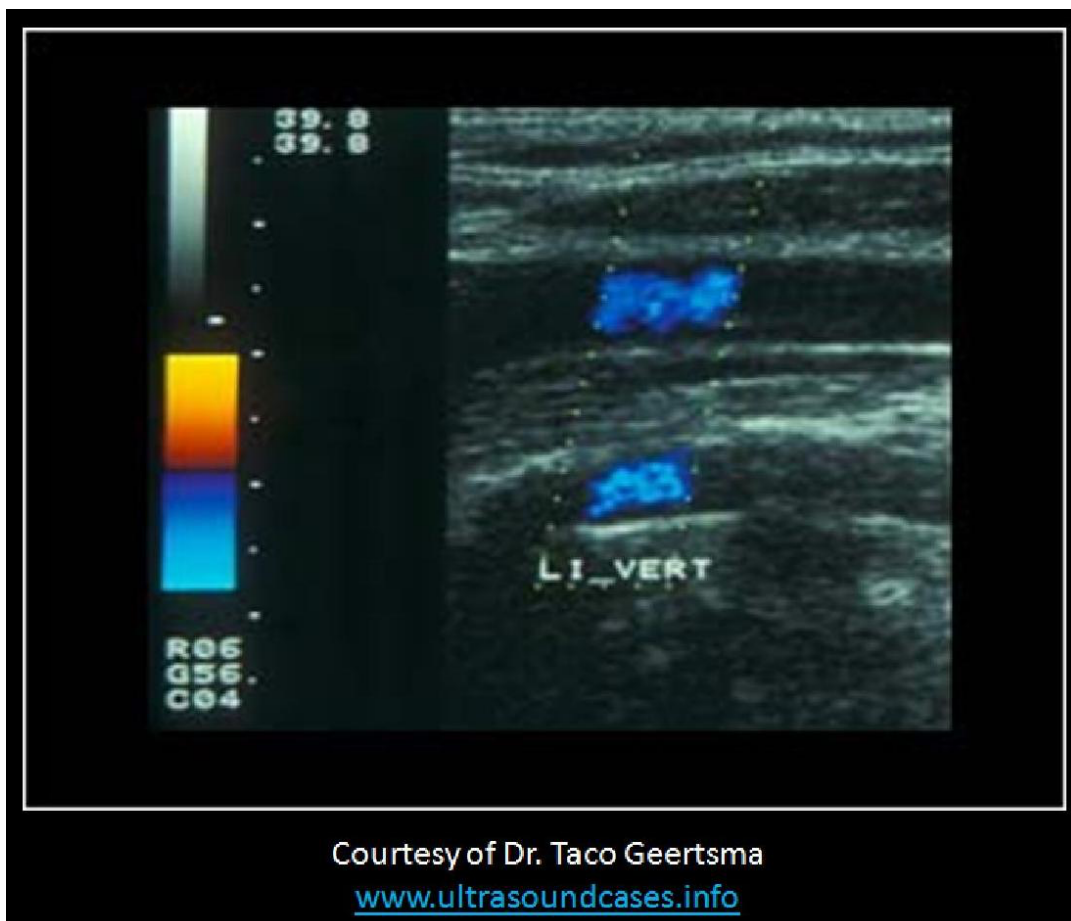
- A. Blood pressure and PPG evaluation of the digital arteries should be performed.
- B. These symptoms are most likely unrelated to arterial disease.
- C. The patient should be evaluated for popliteal entrapment in the left leg.
- D. Hypertensive disease usually leads to overestimation of the ABI values.

150. The soleal plexus is located:

- A. In the forearm
- B. In the neck
- C. In the Circle of Willis of the brain
- D. In the calf

151. What is demonstrated on the image?

- A. normal carotid artery flow with vertebral artery flow reversal
- B. normal carotid artery and vertebral artery flow
- C. normal internal jugular vein flow with normal vertebral artery flow
- D. normal internal jugular vein flow with reversed vertebral artery flow



152. _____ is the gold standard in testing for arterial disease.

- A. Segmental Pressures with Duplex US
- B. Angiography
- C. Angioplasty
- D. Venography

153. A patient presents for follow up after a mesenteric stent placement procedure for a 75% SMA stenosis. Which of the following describes the findings in the SMA that indicate a successful procedure?

- A. Change from monophasic to peaked triphasic
- B. Decreased resistance and increased diastolic flow
- C. Increased resistance and decreased diastolic flow
- D. Change from peaked triphasic to monophasic

154. Which of the following is NOT normally performed after the injection of papaverine into the tissues proximal to the shaft of the penis:

- A. Apply the 2.5cm blood pressure cuff around the shaft of the penis
- B. Doppler waveforms are obtained in the cavernous arteries and veins 1-2 minutes post- injection
- C. Measure the diameter of the cavernous arteries
- D. Repeat the ABI measurements

155. When wearing a protective gown and gloves:

- A. the open end of the gloves should not be touching the wrist cuffs of the gown
- B. two pairs of gloves should be worn
- C. the open end of the gloves should be fitted under the wrist cuffs of the gown
- D. the open end of the gloves should be pulled over the wrist cuffs of the gown

SEQUENCE FOR PUTTING ON PERSONAL PROTECTIVE EQUIPMENT (PPE)	SEQUENCE FOR REMOVING PERSONAL PROTECTIVE EQUIPMENT (PPE)
The type of PPE used will vary based on the level of precautions required, such as standard and contact, droplet or airborne infection isolation precautions. The procedure for putting on and removing PPE should be tailored to the specific type of PPE. 1. GOWN - Fully cover torso from neck to knees, arms to end of wrists, and wrap around the back - Fasten in back of neck and waist  2. MASK OR RESPIRATOR - Secure ties or elastic bands at middle of head and neck - Fit flexible band to nose bridge - Fit snug to face and below chin - Fit-check respirator  3. GOGGLES OR FACE SHIELD - Place over face and eyes and adjust to fit  4. GLOVES - Extend to cover wrist of isolation gown  USE SAFE WORK PRACTICES TO PROTECT YOURSELF AND LIMIT THE SPREAD OF CONTAMINATION - Keep hands away from face - Limit surfaces touched - Change gloves when torn or heavily contaminated - Perform hand hygiene 	Except for respirator, remove PPE at doorway or in anteroom. Remove respirator after leaving patient room and closing door. 1. GLOVES - Outside of gloves is contaminated! - Grasp outside of glove with opposite gloved hand; peel off - Hold removed glove in gloved hand - Slide fingers of ungloved hand under remaining glove at wrist - Peel glove off over first glove - Discard gloves in waste container  2. GOGGLES OR FACE SHIELD - Outside of goggles or face shield is contaminated! - To remove, handle by head band or ear pieces - Place in designated receptacle for reprocessing or in waste container  3. GOWN - Gown front and sleeves are contaminated! - Unfasten ties - Pull away from neck and shoulders, touching inside of gown only - Turn gown inside out - Fold or roll into a bundle and discard  4. MASK OR RESPIRATOR - Front of mask/respirator is contaminated — DO NOT TOUCH! - Grasp bottom, then top ties or elastics and remove - Discard in waste container  PERFORM HAND HYGIENE BETWEEN STEPS IF HANDS BECOME CONTAMINATED AND IMMEDIATELY AFTER REMOVING ALL PPE 

156. A patient presents for a mesenteric Doppler to rule out stenosis. They have a history of diabetes, HTN, pain and nausea post-prandially. Which of the following appointment times would be best for this exam to be scheduled?

- A. 8am, NPO for 6-8hrs prior
- B. 230pm because the patient prefers this time

- C. 1pm, after a good fatty meal
- D. 10am, after light breakfast

157. Which of the following should be increased when evaluating venous flow with color Doppler?

- A. wall filter
- B. color priority
- C. output power
- D. PRF

158. You are performing a mesenteric duplex examination for suspected mesenteric ischemia. The findings are: SMA PSV 408cm/s, EDV 78cm/s, monophasic flow with turbulence Celiac PSV 300cm/s, EDV 55cm/s, monophasic flow with spectral window IMA PSV 180cm/s, EDV 25cm/s, monophasic flow with spectral window How will you report these findings?

- A. suspected mesenteric ischemia with abnormal flow velocities and waveforms in the SMA, IMA, and celiac artery
- B. suspected mesenteric ischemia with abnormal flow velocities and waveforms in the SMA and celiac artery
- C. isolated stenosis of the SMA with compensatory flow in the celiac artery and IMA
- D. isolated stenosis of the celiac artery with compensatory flow in the SMA and IMA

159. A patient presents for an ABI exam due to bilateral calf pain. He is a dialysis patient with an occluded graft in the right arm and a patent graft in the left arm. What should you do first?

- A. Skip the brachial pressures and just do the ankle pressures
- B. Change the exam to a lower extremity arterial exam and start scanning
- C. Take the pressure in the arm with the occluded graft
- D. Call the referring physician to explain that the exam is not possible for this patient

160. You are about to perform a lower extremity arterial duplex on a patient that recently had an MRA run- off exam. According to the attached image, what vessel will demonstrate reduced or absent flow in the left leg?

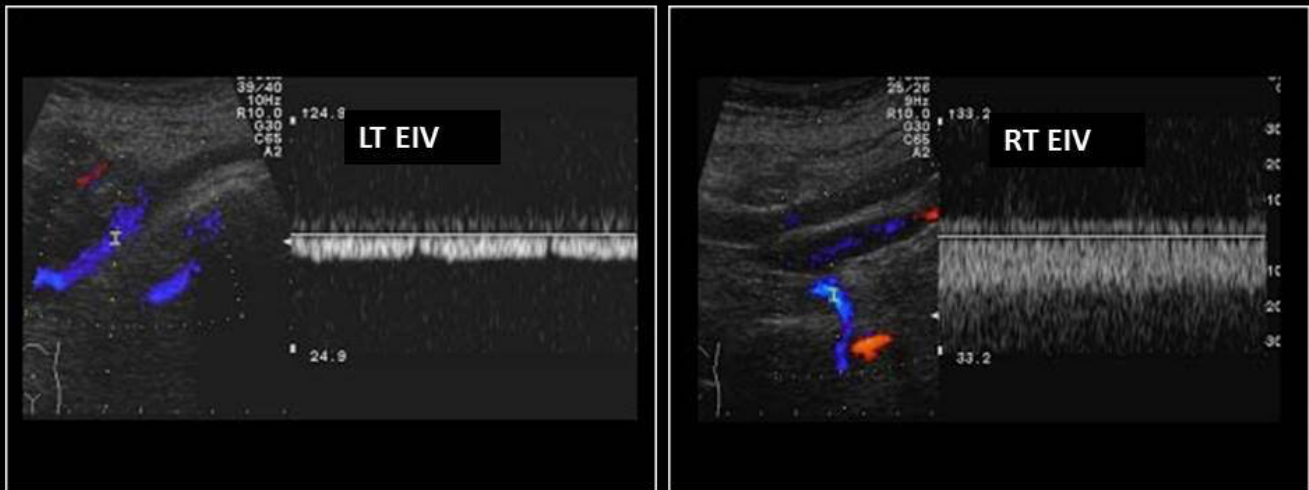
- A. common femoral artery
- B. superficial femoral artery
- C. anterior tibial artery
- D. posterior tibial artery

161. The following pressure readings from a segmental pressure evaluation were obtained in a patient suffering from low back and thigh pain. What is the ABI for each leg? Lt Ankle 106 Rt Ankle 98 Lt Arm 124 Rt Arm 134

- A. Lt = 0.85, Rt = 0.73
- B. Lt = 0.85, Rt = 0.79
- C. Lt = 0.79, Rt = 0.73
- D. Lt = 1.17, Rt = 1.36

162. You perform a bilateral venous Doppler due to unilateral swelling and pain in the right leg. Which of the following should be reported on your technologist worksheet?

- A. Probable May-Thurner syndrome
- B. Probable IVC thrombosis
- C. Probable right common iliac vein thrombosis
- D. Probable left common iliac vein thrombosis



Courtesy of: Dr. Taco Geertsma

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163. When a clot is present on a venogram, how does it appear on the image?

- A. similar to a calcification, very echogenic
- B. dark solid area within the vein
- C. "empty" vein with no contrast opacification
- D. complex mass obstructing contrast flow

Note the absence of contrast in several areas and the irregular filling of a normally smooth walled vein.



Al Ghananeem Z, Deshpande A, Sundaresan V, et al. (August 07, 2024) May-Thurner Syndrome: A Rare, Yet Recognized, Cause of Deep Vein Thrombosis. *Cureus* 16(8): e66357. doi:10.7759/cureus.66357

164. Which of the following techniques can be used to assess the presence and timing of reflux in multiple vessels simultaneously?

- A. Real time 2D imaging
- B. color Doppler

- C. CW Doppler
- D. PW Doppler

165. Where is the most common site of stenosis in a transjugular intrahepatic portosystemic shunt?

- A. hepatic artery anastomosis
- B. portal vein anastomosis
- C. biliary anastomosis
- D. hepatic vein anastomosis

166. Which of the following is true regarding the initial post-op examination of a transjugular intrahepatic portosystemic shunt placement?

- A. CTA is most commonly performed for the initial evaluation of a TIPS procedure.
- B. Marked ascites is a common finding for the initial post op exam on a TIPS patient.
- C. It should be performed within 30 days following the procedure.
- D. It helps to establish baseline velocities for upcoming follow up exams.

167. Which of the following will lead to more color displayed in real time in the abdominal aorta?

- A. increase sample size
- B. decrease color gain
- C. increase persistence
- D. decrease color threshold

168. Which of the following statements correctly describes an endovenous laser ablation of the GSV?

- A. The physician injects a halo of anesthesia medicine around the GSV
- B. The procedure is considered a success if thrombus fills the GSV from the ankle to the SFJ
- C. The procedure is much more easily performed if the patient already has DVT
- D. Intravascular ultrasound is performed after the procedure to confirm success

169. Which of the following has the greatest effect on resistance to blood flow?

- A. Two different arteries with 60% stenosis supplying the same organ
- B. A 60% stenosis in a single artery
- C. Two consecutive areas of 60% stenosis in a single artery
- D. All scenarios above would have the same variation in resistance

170. Which modality is preferred to diagnose fibromuscular dysplasia in the circle of Willis?

- A. Transcranial Doppler
- B. Angiography
- C. MRI
- D. CTA

Answer Key

1. B
2. A
3. A
4. B
5. B
6. A
7. A
8. A
9. A
10. C
11. A
12. C
13. D
14. C
15. A
16. B
17. C
18. B
19. C
20. A
21. C
22. B
23. D
24. B
25. B
26. A
27. A
28. A
29. D
30. C
31. D
- 32.
33. A
34. A
35. C
36. A
37. B
38. B
39. D
40. D
41. D
42. B
43. A
44. C
45. C
46. A
47. C

48. B
49. C
50. D
51. C
52. B
53. B
54. D
55. C
56. A
57. B
58. A
59. A
60. D
61. C
62. D
63. A
64. D
65. B
66. C
67. D
68. A
69. A
70. D
71. D
72. B
73. A
74. C
75. C
76. A
77. B
78. D
79. D
80. A
81. D
82. D
83. C
84. B
85. A
86. D
87. B
88. B
89. C
90. B
91. B
92. A
93. C
94. A
95. C
96. C

97. D
98. D
99. B
100. B
101. C
102. C
103. A
104. D
105. D
106. A
107. D
108. B
109. A
110. D
111. C
112. C
113. D
114. B
115. B
116. C
117. B
118. C
119. B
120. B
121. C
122. B
123.
124. D
125. A
126. A
127. B
128. A
129. D
130. B
131. C
132. A
133. A
134. C
135. C
136. C
137. A
138. C
139. A
140. B
141. D
142. B
143. C
144. B
145. C

146. D
147. B
148.
149. A
150. D
151. B
152. B
153. C
154. D
155. D
156. A
157. B
158. C
159. D
160. C
161. C
162. C
163. C
164. B
165. D
166. D
167. C
168. A
169. C
170. B